



Cisco Nexus 9336PQ ACI-Mode Switch Hardware Installation Guide

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Audience

This publication is for hardware installers and network administrators who install, configure, and maintain Cisco Nexus switches.

Documentation Conventions

Command descriptions use these conventions:

Convention	Description
bold	Bold text indicates the commands and keywords that you enter literally as shown.
<i>Italic</i>	Italic text indicates arguments for which the user supplies the values.
[x]	Square brackets enclose an optional element (keyword or argument).
[x y]	Square brackets enclosing keywords or arguments separated by a vertical bar indicate an optional choice.
{x y}	Braces enclosing keywords or arguments separated by a vertical bar indicate a required choice.
[x {y z}]	Nested set of square brackets or braces indicate optional or required choices within optional or required elements. Braces and a vertical bar within square brackets indicate a required choice within an optional element.
variable	Indicates a variable for which you supply values, in context where italics cannot be used.

Convention	Description
string	A nonquoted set of characters. Do not use quotation marks around the string or the string will include the quotation marks.

Examples use these conventions:

Convention	Description
<code>screen font</code>	Terminal sessions and information the switch displays are in screen font.
boldface screen font	Information you must enter is in boldface screen font.
<i>italic screen font</i>	Arguments for which you supply values are in italic screen font.
< >	Nonprinting characters, such as passwords, are in angle brackets.
[]	Default responses to system prompts are in square brackets.
!, #	An exclamation point (!) or a pound sign (#) at the beginning of a line of code indicates a comment line.

Related Documentation

Web-Based Documentation

The [APIC documentation landing page](#) includes these documents:

- *Cisco APIC Management Information Model Reference Guide*
- *Cisco APIC Python SDK Documentation*
- *Cisco APIC, AVS, and ACI Virtual Edge Support Matrix*
- *Cisco ACI MIB Support List*

Downloadable Documentation

The [APIC documentation landing page](#) includes these documents:

- *Knowledge Base Articles (KB Articles)*
- *Cisco Application Centric Infrastructure Controller Release Notes*
- *Cisco Application Centric Infrastructure Fundamentals Guide*
- *Cisco APIC Getting Started Guide*
- *Cisco ACI Virtualization Guide*
- *Cisco APIC REST API User Guide*
- *Cisco APIC Command Line Interface User Guide*
- *Cisco APIC Faults, Events, and System Messages Management Guide*

- *Cisco ACI System Messages Reference Guide*
- *Cisco APIC Layer 4 to Layer 7 Services Deployment Guide*
- *Cisco ACI Firmware Management Guide*
- *Cisco ACI Troubleshooting Guide*
- *Cisco ACI Switch Command Reference, NX-OS Release 11.0*
- *Cisco Verified Scalability Guide for Cisco ACI*
- *Cisco ACI MIB Quick Reference*
- *Cisco Nexus CLI to Cisco APIC Mapping Guide*
- *Application Centric Infrastructure Fabric Hardware Installation Guide*

The [ACI-mode hardware installation guides landing page](#) includes these documents:

- *Cisco NX-OS Release Notes for Cisco Nexus 9000 Series ACI-Mode Switches*
- *Cisco Nexus 9000 Series ACI Mode Licensing Guide*
- *Cisco Nexus 93108TX-EX ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 93108TX-FX ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 93120TX ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 93128TX ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 93180LC-EX ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 93180YC-EX ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 93180YC-FX ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 9332PQ ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 9336PQ ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 9372PX and 9372PX-E ACI-Mode Switches Hardware Installation Guide*
- *Cisco Nexus 9372TX and 9372TX-E ACI-Mode Switches Hardware Installation Guide*
- *Cisco Nexus 9396PX ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 9396TX ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 9504 ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 9508 ACI-Mode Switch Hardware Installation Guide*
- *Cisco Nexus 9516 ACI-Mode Switch Hardware Installation Guide*

Cisco Application Centric Infrastructure (ACI) Simulator Documentation

The [Cisco ACI Simulator landing page](#) includes these documents:

- *Cisco ACI Simulator Release Notes*

- *Cisco ACI Simulator Installation Guide*
- *Cisco ACI Simulator Getting Started Guide*

Cisco Nexus 9000 Series Switches Documentation

See the [Cisco Nexus 9000 Series Switches](#) landing page.

Cisco Application Virtual Switch Documentation

See the [Cisco Application Virtual Switch \(AVS\) documentation](#) landing page.

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CHAPTER 1

Overview

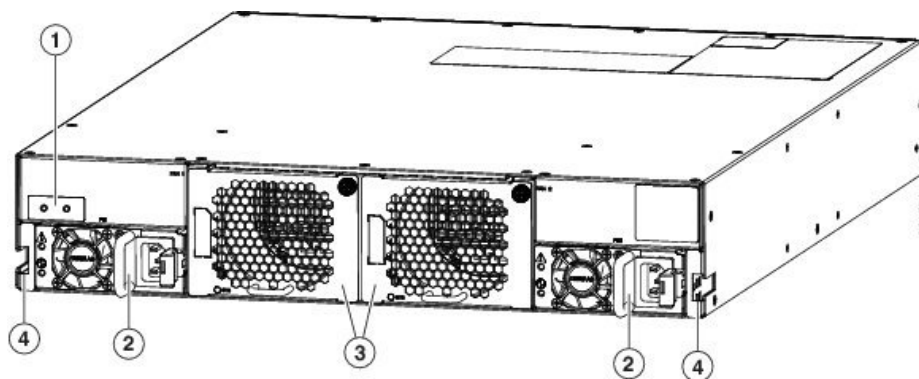
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Overview

The Cisco Nexus 9336PQ ACI Spine Switch (N9K-C9336PQ) is a 2-rack-unit (RU) switch for the Cisco Application-Centric Infrastructure (ACI). This switch includes the following components:

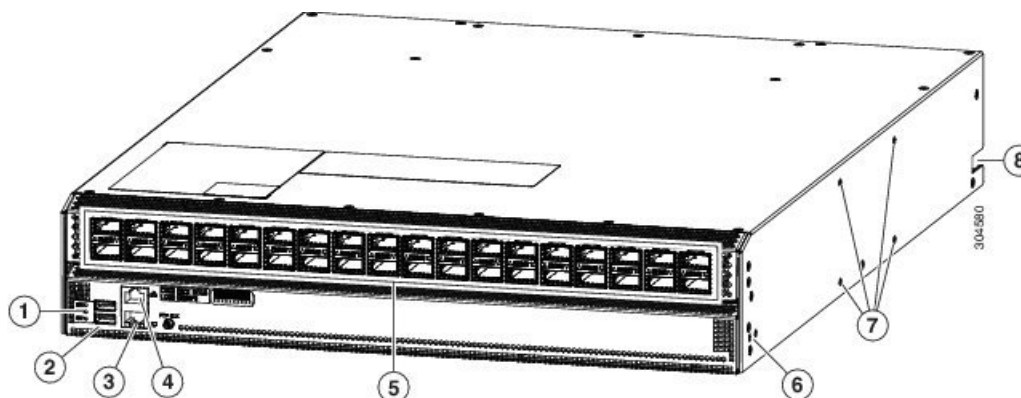
- 36 fixed 40-Gigabit Quad Small Form-factor Pluggable (QSFP+) ports
- Console and out-of-band-management ports (one each)
- Fan modules (two)
 - Port-side intake version with burgundy coloring (N9K-C9300-FAN3)
 - Port-side exhaust version with blue coloring (N9K-C9300-FAN3-B)
- Power supplies (two—one required for operations and an optional one for redundancy)
 - 1200-W AC power supply with port-side intake airflow (burgundy coloring) (N9K-PAC-1200W)
 - 1200-W AC power supply with port-side exhaust airflow (blue coloring) (N9K-PAC-1200W-B)
 - 1200-W HVAC/HVDC power supply with dual-direction airflow (white coloring) (N9K-PUV-1200W)
 - 930-W DC power supply with port-side intake airflow (green coloring) (UCSC-PSU-930WDC)
 - 930-W DC power supply with port-side exhaust airflow (gray coloring) (UCS-PSU-6332-DC)

The following figure shows the hardware features seen on the power supply side of the chassis.



1	Grounding Pad	3	Fan modules (2)
2	Power supplies (2) (AC power supply shown)		

The following figure shows the hardware features seen from the port side of the chassis.



1	Chassis LEDs	5	36 40-Gigabit QSFP+ ports
2	USB ports	6	Screw holes (2) for attaching a front-mount bracket for 4-post racks (1 bracket on each of 2 sides)
3	Out-of-band management port (RJ-45 port)	7	Notch in both sides of the chassis for locking the power supply end of the chassis to the bottom support brackets
4	Console port (RJ232 port)		

Ports

The switch chassis includes the following types of ports:

- 40-Gigabit QSFP+ interface ports (36)
- Console port (1)

- Out-of-band management port (1)
- USB flash-memory ports (2)

The interface ports support QSFP+ transceivers. You use these ports to connect the switch to leaf switches in the ACI network that this switch belongs to. These ports are numbered 1 to 36 from left to right.



Note To see which transceivers are supported by this switch, see <http://www.cisco.com/c/en/us/support/interfaces-modules/transceiver-modules/products-device-support-tables-list.html>.

The console port enables you to perform the initial setup of the switch.

The out-of-band management port enables you to manage switch operations after you set up the switch.

If you need to download or upload switch configurations on a flash drive, you can use the flash drive in one of the two USB ports.

Fan Modules

The switch supports two fan modules, which provide port-side intake or port-side exhaust airflow for cooling. You can hot swap one of the fan modules during operations but must replace it within one minute. If you cannot replace the fan module within one minute, then it is best to leave the failed fan module in the chassis until you are ready to replace it.

The fan modules are labeled FAN 1 on the left and FAN 2 on the right.



Note The switch must run with all of its power supply and fan modules taking in cooling air from a cold aisle and exhausting to a hot aisle. If they take in air from the hot aisle, the switch can overheat and shut down. All fan and power supply modules must use the same direction of airflow.

Power Supply Modules

The switch has two power supply slots labeled PS1 on the left and PS2 on the right. You can mix any of the following power supplies in those slots:

- 1200-W port-side intake AC power supply with burgundy coloring (N9K-PAC-1200W)
- 1200-W port-side exhaust AC power supply with blue coloring (N9K-PAC-1200W-B)
- 930-W DC power supplies with green coloring for port-side intake airflow (UCSC-PSU-930WDC)
- 930-W DC power supplies with gray coloring for port-side exhaust airflow (UCS-PSU-6332-DC)

The switch requires one power supply for its operations and a second power supply for power redundancy. You can hot swap one of the power supplies during operations but otherwise must always have two power supplies (or one power supply and one blank faceplate) installed in the chassis to maintain the designed airflow.

**Caution**

The switch must run with all of its power supply and fan modules taking in cooling air from a cold aisle and exhausting to a hot aisle. If they take in air from the hot aisle, the switch can overheat and shut down. All power supply and fan modules must use the same direction of airflow.



CHAPTER 2

Preparing the Site

- [Temperature Requirements, on page 5](#)
- [Humidity Requirements, on page 6](#)
- [Altitude Requirements, on page 6](#)
- [Dust and Particulate Requirements, on page 6](#)
- [Minimizing Electromagnetic and Radio Frequency Interference, on page 7](#)
- [Shock and Vibration Requirements, on page 8](#)
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- [Clearance Requirements, on page 11](#)
- [Network Equipment-Building System \(NEBS\) Statements, on page 12](#)

Temperature Requirements

The operating temperature of the switch is 32 to 104 degrees Fahrenheit (0 to 40 degrees Celsius) at sea level. For every 300 meters (1000 feet) above sea level, the maximum temperature is reduced by 1 degree Celsius. The non-operating temperature of the switch is -40 to 158 degrees Fahrenheit (-40 to 70 degrees Celsius).

Overview of Module Temperatures

Built-in, automatic sensors in all switches in the Cisco Nexus 9000 Series monitor your switch at all times. Each module (supervisor, I/O, and fabric) has temperature sensors with two thresholds:



Note For any Major temperature alarms from the sensors, the switch powers down in 2 minutes. Power on the switch after fixing the temperature issue.

- Minor temperature threshold—If exceeded, a minor alarm occurs and these actions happen for all four sensors:
 - System messages display.
 - System sends Call Home alerts (if configured).

- System sends SNMP notifications (if configured).
 - System fan speed will increment.
- Major temperature threshold—If exceeded, a major alarm occurs and these actions happen:
- If the threshold is exceeded in a switching module, only that module is shut down.
- For all sensors:
- System messages display.
 - System sends Call Home alerts (if configured).
 - System sends SNMP notifications (if configured).
 - System fan speed will increment.
 - If the major threshold is exceeded in a switching module, only that module is shut down.
 - If the major threshold is exceeded in an active supervisor module with HA-standby or standby present, only that supervisor module is shut down and the standby supervisor module takes over.
 - If you do not have a standby supervisor module in your switch, you have 2 minutes to decrease the temperature. During this interval, the software monitors the temperature every 5 seconds and continuously sends system messages every 10 seconds, as configured.

Humidity Requirements

High humidity can cause moisture to enter the switch. Moisture can cause corrosion of internal components and degradation of properties such as electrical resistance, thermal conductivity, physical strength, and size. The switch is rated to withstand from 5- to 95-percent (nonoperating) and 5- to 90-percent (operating) relative humidity.

Climate-controlled buildings usually maintain an acceptable level of humidity for the switch equipment. If the switch is located in an unusually humid location, use a dehumidifier to maintain the humidity within an acceptable range.

Altitude Requirements

Altitude rating is 10,000 ft (3048 m). For China, it is 6,562 ft (2000 m).

Dust and Particulate Requirements

Exhaust fans cool power supplies. System fans cool switches by drawing in air and exhausting air out through various openings in the chassis. Fans also introduce dust and other particles, causing contaminant buildup in the switch and increased internal chassis temperature. Dust and particles can act as insulators and interfere with the mechanical components in the switch. Keep a clean operating environment to reduce the negative effects of dust and other particles.

In addition to keeping your environment free of dust and particles, use these precautions to avoid contamination of your switch:

- Do not smoke near the switch.
- Do not eat or drink near the switch.

Minimizing Electromagnetic and Radio Frequency Interference

Electromagnetic interference (EMI) and radio frequency interference (RFI) from the switch can adversely affect other devices, such as radio and television (TV) receivers. Radio frequencies that emanate from the switch can also interfere with cordless and low-power telephones. Conversely, RFI from high-power telephones can cause spurious characters to appear on the switch monitor.

RFI is defined as any EMI with a frequency above 10 kHz. This type of interference can travel from the switch to other devices through the power cable and power source or through the air as transmitted radio waves. The Federal Communications Commission (FCC) publishes specific regulations to limit the amount of EMI and RFI that are emitted by computing equipment. Each switch meets these FCC regulations.

To reduce the possibility of EMI and RFI, use these guidelines:

- Cover all open expansion slots with a blank filler plate.
- Always use shielded cables with metal connector shells for attaching peripherals to the switch.

When wires are run for any significant distance in an electromagnetic field, interference can occur to the signals on the wires with these implications:

- Bad wiring can result in radio interference emanating from the plant wiring.
- Strong EMI, especially when it is caused by lightning or radio transmitters, can destroy the signal drivers and receivers in the chassis and even create an electrical hazard by conducting power surges through lines into equipment.



Note To predict and prevent strong EMI, consult experts in radio frequency interference (RFI).

The wiring is unlikely to emit radio interference if you use a twisted-pair cable with a good distribution of grounding conductors. Copper cables should not be longer than maximum distances for the media type.



Caution If the wires exceed the recommended distances, or if wires pass between buildings, give special consideration to the effect of a lightning strike in your vicinity. The electromagnetic pulse that is caused by lightning or other high-energy phenomena can easily couple enough energy into unshielded conductors to destroy electronic switches. Consult experts in electrical surge suppression and shielding if you have had similar problems in the past.

Shock and Vibration Requirements

The switch has been shock- and vibration-tested for operating ranges, handling, and earthquake standards.

Preventing Electrostatic Discharge Damage

Many switch components can be damaged by static electricity. Not exercising the proper electrostatic discharge (ESD) precautions can result in intermittent or complete component failures. To minimize the potential for ESD damage, always use an ESD-preventive anti-static wrist strap (or ankle strap) and ensure that it makes adequate skin contact.



Note Check the resistance value of the ESD-preventive strap periodically. The measurement should be 1–10 megohms. Before you perform any of the procedures in this guide, attach an ESD-preventive strap to your wrist and connect the leash to the chassis.

Grounding Requirements

The switch is sensitive to variations in voltage that is supplied by the power sources. Overvoltage, undervoltage, and transients (spikes) can erase data from memory or cause components to fail. To protect against these types of problems, ensure that there is an earth-ground connection for the switch.

Connect the grounding pad on the switch either directly to the earth-ground connection or to a fully bonded and grounded rack.

When the chassis is properly installed in a grounded rack, the switch is grounded because it has a metal-to-metal (no paint, stain, dirt, or anything else on it) contact to the rack. See Note to ensure proper conductivity between rack and switch is maintained.

Alternatively, ground the chassis by using a customer-supplied grounding cable that meets your local and national installation requirements. For U.S. installations, we recommend 6-AWG wire. Connect your grounding cable to the chassis with a grounding lug (provided in the switch accessory kit) and to the facility ground.



Note Create an electrical conducting path between the product chassis and the metal surface of the enclosure, or rack in which it is mounted, or to a grounding conductor. Provide electrical continuity by using thread-forming type mounting screws that remove any paint or non-conductive coatings and establish a metal-to-metal contact. Remove any paint or other non-conductive coatings on the surfaces between the mounting hardware and the enclosure or rack. Clean the surfaces and apply an antioxidant before installation.

Planning for Power Requirements

The switch includes two power supplies (1-to-1 redundancy with current sharing) in one of these combinations:

- Two 1200-W AC power supplies

- Two 1200-W HVAC/HVDC power supplies
- Two 930-W DC power supplies



Note Both power supplies must be the same type. Do not mix AC, DC, and HVAC/HVDC power supplies in the same chassis.



Note For 1+1 redundancy, you must use two power sources and connect each power supply to a separate power source.

The power supplies are rated to output up to 1200 W (AC power supplies) or up to 930 W (DC power supplies), but the switch requires less than those amounts of power from the power supply. To operate the switch, you must provision enough power from the power source to cover the requirements of both the switch and a power supply. Typically, this switch and a power supply require about 400 W of power input from the power source, but you must provision as much as 660 W power input from the power source to cover peak demand.



Note Some of the power supply modules have rating capabilities that exceed the switch requirements. When calculating your power requirements, use the switch requirements to determine the amount of power that is required for the power supplies.



Note In any power redundancy mode, the total power is shared across the populated Power Supply Units (PSUs). Multiple PSUs are always load-sharing regardless of the mode used. If a PSU has an active input power, the PSU provides a fraction of the total power to the system. This ensures that all active power supplies contribute to the power load, sharing it proportionally to maintain power redundancy.

To minimize the possibility of circuit failure, verify that each power-source circuit that is used by the switch is dedicated to the switch.



Warning **Statement 1005**—Circuit Breaker when using DC power supplies

This product relies on the building's installation for short-circuit (overcurrent) protection.

- Ensure that the protective devices are rated not greater than 30A when the switch is powered with regular DC power supplies (rated 48-60VDC).
- Ensure that the protective devices are rated not greater than 10A when the switch is powered with HVDC power supplies (rated 240-350VDC).

Airflow Requirements

The switch is positioned with its ports in either the front or the rear of the rack, depending on your cabling and maintenance requirements. To identify the airflow options for your switch, see the user-replaceable components in the *Overview* section of this document. Position the fan and power supply modules to move the coolant air from the cold aisle to the hot aisle in one of these ways:

- Port-side exhaust airflow—Cool air enters the chassis through the fan and power supply modules in the cold aisle and exhausts through the port end of the chassis in the hot aisle.
- Port-side intake airflow—Cool air enters the chassis through the port end in the cold aisle and exhausts through the fan and power supply modules in the hot aisle.
- Dual-direction airflow—The direction of the installed fan modules determines the airflow.
- Single-direction airflow—The direction of the installed fan modules determines the airflow.

Identify the airflow direction of each fan and power supply module by its coloring.

- Gray coloring on DC power supplies indicates port-side exhaust airflow.
- Green coloring on DC power supplies indicates port-side intake airflow.



Note To prevent the switch from overheating and shutting down, position the air intake for the switch in a cold aisle. The fan and power supply modules must have the same direction of airflow. To change the airflow direction for the switch, shutdown the switch before changing the modules.

Airflow Requirements

The switch is positioned with its ports in either the front or the rear of the rack depending, on your cabling and maintenance requirements. Position the fan and power supply modules to move the coolant air from the cold aisle to the hot aisle in one of these ways:

- Port-side exhaust airflow—Coolant air enters the chassis through the fan and power supply modules in the cold aisle and exhausts through the port end of the chassis in the hot aisle.
- Port-side intake airflow—Coolant air enters the chassis through the port end in the cold aisle and exhausts through the fan and power supply modules in the hot aisle.
- Dual-direction airflow—The direction of the installed fan modules determines the airflow.

You can identify the airflow direction of each fan and power supply module by its coloring as follows:

- Blue coloring indicates port-side exhaust airflow.
- Burgundy coloring indicates port-side intake airflow.
- White coloring on HVAC/HVDC power supplies indicates dual direction airflow.
- Gray coloring on DC power supplies indicates port-side exhaust airflow.

- Green coloring on DC power supplies indicates port-side intake airflow.

**Note**

To prevent the switch from overheating and shutting down, you must position the air intake for the switch in a cold aisle. The fan and power supply modules must have the same direction of airflow (even if their coloring is different). If you must change the airflow direction for the switch, you must shutdown the switch before changing the modules.

Rack and Cabinet Requirements

Install these types of racks or cabinets for your switch:

- Standard perforated cabinets
- Solid-walled cabinets with a roof fan tray (bottom-to-top cooling)
- Standard open four-post Telco racks

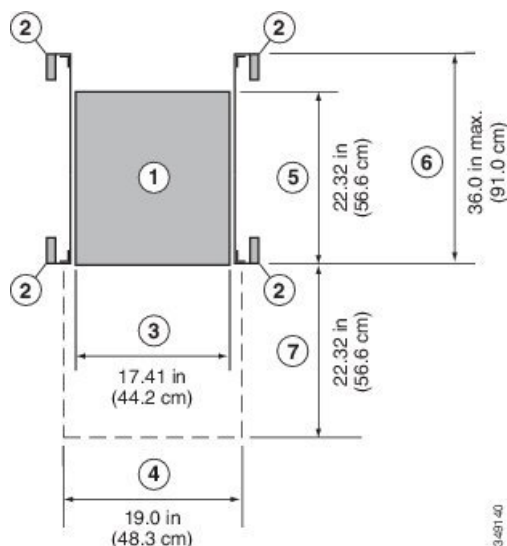
Work with your cabinet vendors to determine which of their cabinets meet these requirements or see the Cisco Technical Assistance Center (TAC) for recommendations:

- Use a standard 19-inch (48.3-cm), four-post Electronic Industries Alliance (EIA) cabinet or rack with mounting rails that conform to English universal hole spacing per section 1 of the ANSI/EIA-310-D-1992 standard.
- The depth of a four-post rack must be 24 to 32 inches (61.0 to 81.3 cm) between the front and rear mounting rails (for proper mounting of the bottom-support brackets or other mounting hardware).
- Required clearances between the chassis and the edges of its rack or the interior of its cabinet are:
 - 4.5 inches (11.4 cm) between the front of the chassis and the interior of the cabinet (required for cabling).
 - 3.0 inches (7.6 cm) between the rear of the chassis and the interior of the cabinet (required for airflow in the cabinet if used).
 - No clearance is required between the chassis and the sides of the rack or cabinet (no side airflow).

Also, you must have power receptacles that are located within reach of the power cords that are used with the switch.

Clearance Requirements

Provide the chassis with adequate clearance between the chassis and any other rack, device, or structure so that you can properly install the switch. Provide the chassis with adequate clearance to route cables, provide airflow, and maintain the switch. For the clearances required for an installation of this chassis in a four-post rack, see the figure.



1	Chassis	5	Depth of the chassis
2	Vertical rack-mount posts and rails	6	Maximum extension of the bottom-support rails 36.0 in (91.4 cm)
3	Chassis width 17.3 in (43.9 cm)	7	Depth of the front clearance area (equal to the depth of the chassis).
4	Width of the front clearance area (equal to the width of the chassis with two rack-mount brackets that are attached to it). 19.0 in (48.3 cm)		



Note Both the front and rear of the chassis must be open to both aisles for airflow.

Network Equipment-Building System (NEBS) Statements

The regulatory compliance statements and requirements for the Network Equipment Building System (NEBS) certification are listed here.



Note **Statement 7001**—ESD Mitigation

This equipment may be ESD sensitive. Always use an ESD ankle or wrist strap before handling equipment. Connect the equipment end of the ESD strap to an unfinished surface of the equipment chassis or to the ESD jack on the equipment if provided.

**Warning****Statement 7003—Shielded Cable Requirements for Intrabuilding Lightning Surge**

The intrabuilding port(s) of the equipment or subassembly must use shielded intrabuilding cabling/wiring that is grounded at both ends.

The following port(s) are considered intrabuilding ports on this equipment:

**Note****Statement 7004—Special Accessories Required to Comply with GR-1089 Emission and Immunity Requirements**

To comply with the emission and immunity requirements of GR-1089, shielded cables are required for the following ports:

**Warning****Statement 7005—Intrabuilding Lightning Surge and AC Power Fault**

The intrabuilding port(s) of the equipment or subassembly is suitable for connection to intrabuilding or unexposed wiring or cabling only. The intrabuilding port(s) of the equipment or subassembly **MUST NOT** be metalically connected to interfaces that connect to the OSP or its wiring for more than 6 meters (approximately 20 feet). These interfaces are designed for use as intrabuilding interfaces only (Type 2, 4, or 4a ports as described in GR-1089) and require isolation from the exposed OSP cabling. The addition of primary protectors is not sufficient protection in order to connect these interfaces metalically to an OSP wiring system.

The following ports are considered intrabuilding ports on the equipment:

**Warning****Statement 7012—Equipment Interfacing with AC Power Ports**

Connect this equipment to AC mains that are provided with a surge protective device (SPD) at the service equipment that complies with NFPA 70, the National Electrical Code (NEC).

**Note****Statement 7013—Equipment Grounding Systems—Common Bonding Network (CBN)**

This equipment is suitable for installations using the CBN.

**Note****Statement 7015—Equipment Bonding and Grounding**

When you use thread-forming screws to bond equipment to its mounting metalwork, remove any paint and nonconductive coatings and clean the joining surfaces. Apply an antioxidant compound before joining the surfaces between the equipment and mounting metalwork.

**Note** **Statement 7016**—Battery Return Conductor

Treat the battery return conductor of this equipment as

**Note** **Statement 7018**—System Recover Time

The equipment is designed to boot up in less than 30 minutes provided the neighboring devices are fully operational.

**Note** **Statement 8015**—Installation Location Network Telecommunications Facilities

This equipment is suitable for installation in network telecommunications facilities.

**Note** **Statement 8016**—Installation Location Where the National Electric Code (NEC) Applies

This equipment is suitable for installation in locations where the NEC applies.

**Warning** **Statement 1056**—Unterminated Fiber Cable

Invisible laser radiation may be emitted from the end of the unterminated fiber cable or connector. Do not view directly with optical instruments. Viewing the laser output with certain optical instruments, for example, eye loupes, magnifiers, and microscopes, within a distance of 100 mm, may pose an eye hazard.

**Warning** **Statement 1255**—Laser Compliance Statement

Pluggable optical modules comply with IEC 60825-1 Ed. 3 and 21 CFR 1040.10 and 1040.11 with or without exception for conformance with IEC 60825-1 Ed. 3 as described in Laser Notice No. 56, dated May 8, 2019.



CHAPTER 3

Installing the Chassis

- [Safety, on page 15](#)
- [Installation Options with Rack-Mount Kits, on page 18](#)
- [Install a Rack, on page 19](#)
- [Unpacking and Inspecting a New Switch, on page 20](#)
- [Planning How to Position the Chassis in the Rack, on page 21](#)
- [Installing the Chassis in a Four-Post Rack, on page 21](#)
- [Grounding the Chassis, on page 27](#)
- [Start the switch, on page 29](#)

Safety

Before you install, operate, or service the switch, see the *Regulatory, Compliance, and Safety Information for the Cisco Nexus 9000 Series* content for important Safety Information.



Warning

Statement 1071—Warning Definition

IMPORTANT SAFETY INSTRUCTIONS

Before you work on any equipment, be aware of the hazards involved with electrical circuitry and be familiar with standard practices for preventing accidents. Read the installation instructions before using, installing, or connecting the system to the power source. Use the statement number at the beginning of each warning statement to locate its translation in the translated safety warnings for this device.

SAVE THESE INSTRUCTIONS





Note **Statement 1089**—Instructed and Skilled Person Definitions

An instructed person is someone who has been instructed and trained by a skilled person and takes the necessary precautions when working with equipment.

A skilled person or qualified personnel is someone who has training or experience in the equipment technology and understands potential hazards when working with equipment.



Warning **Statement 1074**—Comply with Local and National Electrical Codes

To reduce risk of electric shock or fire, installation of the equipment must comply with local and national electrical codes.



Note **Statement 407**—Japanese Safety Instruction

You are strongly advised to read the safety instruction before using the product.

<https://www.cisco.com/web/JP/techdoc/pldoc/pldoc.html>

When installing the product, use the provided or designated connection cables/power cables/AC adapters.

〈製品使用における安全上の注意〉

www.cisco.com/web/JP/techdoc/index.html

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Warning **Statement 1017**—Restricted Area

This unit is intended for installation in restricted access areas. Only skilled, instructed, or qualified personnel can access a restricted access area.

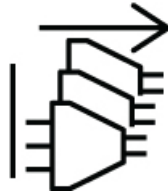


Warning **Statement 1091**—Installation by an Instructed Person or Skilled Person

Only an instructed person or skilled person should be allowed to install, replace, or service this equipment. See statement 1089 for the definition of an instructed person or skilled person.

**Warning****Statement 1028—More Than One Power Supply**

This unit might have more than one power supply connection. To reduce risk of electric shock, remove all connections to de-energize the unit.

**Warning****Statement 1003—DC Power Disconnection**

To reduce risk of electric shock or personal injury, disconnect DC power before removing or replacing components or performing upgrades.

**Warning****Statement 1046—Installing or Replacing the Unit**

To reduce risk of electric shock, when installing or replacing the unit, the ground connection must always be made first and disconnected last.

If your unit has modules, secure them with the provided screws.

**Warning****Statement 1022—Disconnect Device**

To reduce the risk of electric shock and fire, a readily accessible disconnect device must be incorporated in the fixed wiring.

**Warning****Statement 1033—Safety Extra-Low Voltage (SELV)—IEC 60950/ES1–IEC 62368 DC Power Supply**

To reduce the risk of electric shock, connect the unit *only* to a DC power source that complies with the SELV requirements in the IEC 60950-based safety standards or the ES1 requirements in the IEC 62368-based safety standards.

**Warning****Statement 1024—Ground Conductor**

This equipment must be grounded. To reduce the risk of electric shock, never defeat the ground conductor or operate the equipment in the absence of a suitably installed ground conductor. Contact the appropriate electrical inspection authority or an electrician if you are uncertain that suitable grounding is available.

**Warning** **Statement 1032—Lifting the Chassis**

To prevent personal injury or damage to the chassis, never attempt to lift or tilt the chassis using the handles on modules, such as power supplies, fans, or cards. These types of handles are not designed to support the weight of the unit.

**Warning** **Statement 1006—Chassis Warning for Rack-Mounting and Servicing**

To prevent bodily injury when mounting or servicing this unit in a rack, you must take special precautions to ensure that the system remains stable. The following guidelines are provided to ensure your safety:

- This unit should be mounted at the bottom of the rack if it is the only unit in the rack.
- When mounting this unit in a partially filled rack, load the rack from the bottom to the top with the heaviest component at the bottom of the rack.
- If the rack is provided with stabilizing devices, install the stabilizers before mounting or servicing the unit in the rack.

**Warning** **Statement 1056—Unterminated Fiber Cable**

Invisible laser radiation may be emitted from the end of the unterminated fiber cable or connector. Do not view directly with optical instruments. Viewing the laser output with certain optical instruments, for example, eye loupes, magnifiers, and microscopes, within a distance of 100 mm, may pose an eye hazard.



Caution To prevent loss of input power, ensure the total maximum loads on the circuits supplying power to the switch are within the current ratings for the wiring and breakers.



Note For AC input application, please refer to the statement below:

**Warning** **Statement 1005—Circuit Breaker**

This product relies on the building's installation for short-circuit (overcurrent) protection. Ensure that the protective devices is rated not greater than 20A (North America), 16A (Europe), and 13A (UK).

Installation Options with Rack-Mount Kits

The rack-mount kit enables you to install the switch into racks of varying depths. Position the switch with easy access to either the port connections or the fan and power supply modules.

Install the switch using these rack-mount options:

- Rack-mount kit (N9K-C9300-RMK) which you can order from Cisco.

For four post racks.

The rack or cabinet that you use must meet the requirements listed the in [General Requirements for Cabinets and Racks, on page 47](#) section.



Note You are responsible for verifying that your rack and rack-mount hardware comply with the guidelines that are described in this document.

Install a Rack

Before you install the switch, you must install a standard four-post, 19-inch EIA data center rack (or a cabinet that contains such a rack) that meets the requirements listed in [Overview of Racks, on page 47](#).

Procedure

Step 1 Bolt the rack to the concrete subfloor before moving the chassis onto it.

Warning

Statement 1048—Rack Stabilization

The rack stabilizing mechanism must be in place, or the rack must be bolted to the floor before installation or servicing. Failure to stabilize the rack can cause bodily injury.

Step 2 If the rack has bonded construction, connect it to the earth ground. This action enables you to easily ground the switch and its components and to ground your electrostatic discharge (ESD) wrist strap to prevent damaging discharges when you handle ungrounded components before installing them.

Step 3 Include one or two power sources at the rack. For AC power, provide a power receptacle.

Step 4 Include one or two power sources at the rack. For DC power, provide a circuit breaker with terminals for connecting power cables.

Warning

Statement 1018—Supply Circuit

To reduce risk of electric shock and fire, take care when connecting units to the supply circuit so that wiring is not overloaded.

Note

If you are not using power redundancy or are using $n+1$ redundancy, you need only one power source. If you are using $n+n$ redundancy, you need two power sources.

Unpacking and Inspecting a New Switch

Before you install a new chassis, unpack and inspect it to be sure that you have all the items that you ordered. Verify that the switch was not damaged during shipment. If anything is damaged or missing, contact your customer representative immediately.



Caution

When you handle the chassis or its components, follow ESD protocol at all times to prevent ESD damage. This protocol includes but is not limited to wearing an ESD wrist strap that you connect to the earth ground.



Tip

Do not discard the shipping container when you unpack the switch. Flatten the shipping cartons and store them. If you need to move or ship the system in the future, you will need this container.

Procedure

Step 1 Compare the shipment to the equipment list that is provided by your customer service representative. Verify that you have received all of the ordered items.

The shipment should include:

- Switch chassis, which includes these installed components:
 - Two power supplies (any combination of these, with the airflow direction being the same as for the fan modules):
 - 1200-W AC power supply
 - Port-side exhaust airflow with blue coloring (N9K-PAC-1200W-B)
 - Port-side intake airflow with burgundy coloring (N9K-PAC-1200W)
 - 930-W DC power supply
 - Port-side exhaust airflow with gray coloring (UCS-PSU-6332-DC)
 - Port-side intake airflow with green coloring (UCSC-PDU-930WDC)
 - Two fan modules (all fan and power supply modules must have the same airflow direction)
 - Port-side exhaust airflow with blue coloring (N9K-C9300-FAN3-B)
 - Port-side intake airflow with burgundy coloring (N9K-C9300-FAN3)
- Switch accessory kit

Step 2 Check the contents of the box for damage.

Step 3 If you notice any discrepancies or damage, send this information to your customer service representative by email:

- Invoice number of the shipper (see the packing slip)

- Model and serial number of the missing or damaged unit
 - Description of the problem and how it affects the installation
-

Planning How to Position the Chassis in the Rack

The switch is designed so that you can have coolant air flow through the switch in one of the two following directions:

- Enter the port side and exhaust out the power supply side (port-side intake airflow)
- Enter the power supply side and exhaust out the port side (port-side exhaust airflow)

For port-side intake airflow, the switch must have port-side intake fan and AC power supply modules with one or more of the following colorings:

- Burgundy coloring on fan modules and AC power supplies

For port-side exhaust airflow, the switch must have port-side exhaust fan and AC power supply modules with one or more of the following colorings:

- Blue coloring on fan modules and AC power supplies

You can plan the positioning of the switch so that its ports are located close to ports on connected devices or so that the fan and power supply modules are conveniently located in a maintenance aisle, and then order the modules that move coolant air in the appropriate direction from the cold aisle to the hot aisle.



Note All fan and power supply modules in the same switch must operate with the same direction of airflow and the air intake portion of the switch must be located in a cold aisle.

Installing the Chassis in a Four-Post Rack

Before you install the chassis, be sure that the rack is fully secured to the data center floor.

The table lists the items contained in the rack-mount kit.

Installing the Switch using the N9K-C9300-RMK Rack-mount Kit

Attaching the Bottom-Support Rails to the Rack

The switch chassis that you are installing ships with two adjustable bottom-support rails that you can attach to a four-post rack to hold the chassis. Each of these bottom-support rails has two pieces—one that slides into the other so that you can adjust them to fit racks with front and rear mounting posts that are spaced less than 36 inches (91 cm). On each bottom-support rail, the rail half that slides into the other rail includes a chassis stop that fits into the module end of the chassis. Depending on direction of the chassis airflow, position the

rail half with the chassis stop so that the fan and power supply modules end up in the appropriate aisle as follows:

- Port-side intake (burgundy coloring for fan modules) airflow requires that the bottom-support rail with the chassis stop be located on the hot aisle side of the rack.
- Port-side exhaust (blue coloring for fan modules) airflow requires that the bottom-support rail with the chassis stop be located on the cold aisle side of the rack.



Warning **Statement 1074**—Comply with Local and National Electrical Codes

To reduce risk of electric shock or fire, installation of the equipment must comply with local and national electrical codes.

Before you begin

Before you can install the bottom support rails for the chassis, do this:

- Verify that a four-post rack or cabinet is installed.
- If any other devices are stored in the rack or cabinet, verify that the heavier switches are installed below lighter switches.
- Verify that the bottom-support rails kit is included in the switch accessory kit.
- Verify that you have 8 screws for attaching the bottom support brackets to the racks (typically M6 x 10 mm screws or the screw appropriate for the vertical mounting rails on the rack).



Warning **Statement 1006**—Chassis Warning for Rack-Mounting and Servicing

To prevent bodily injury when mounting or servicing this unit in a rack, you must take special precautions to ensure that the system remains stable. The following guidelines are provided to ensure your safety:

- This unit should be mounted at the bottom of the rack if it is the only unit in the rack.
 - When mounting this unit in a partially filled rack, load the rack from the bottom to the top with the heaviest component at the bottom of the rack.
 - If the rack is provided with stabilizing devices, install the stabilizers before mounting or servicing the unit in the rack.
-

Procedure

Step 1 Look at the fan and power supply modules installed in the chassis to determine how you must position the bottom-support rails on the rack.

- If the fan modules have blue coloring (port-side exhaust modules), position the bottom support rails so that the chassis stop is positioned by the cold aisle.

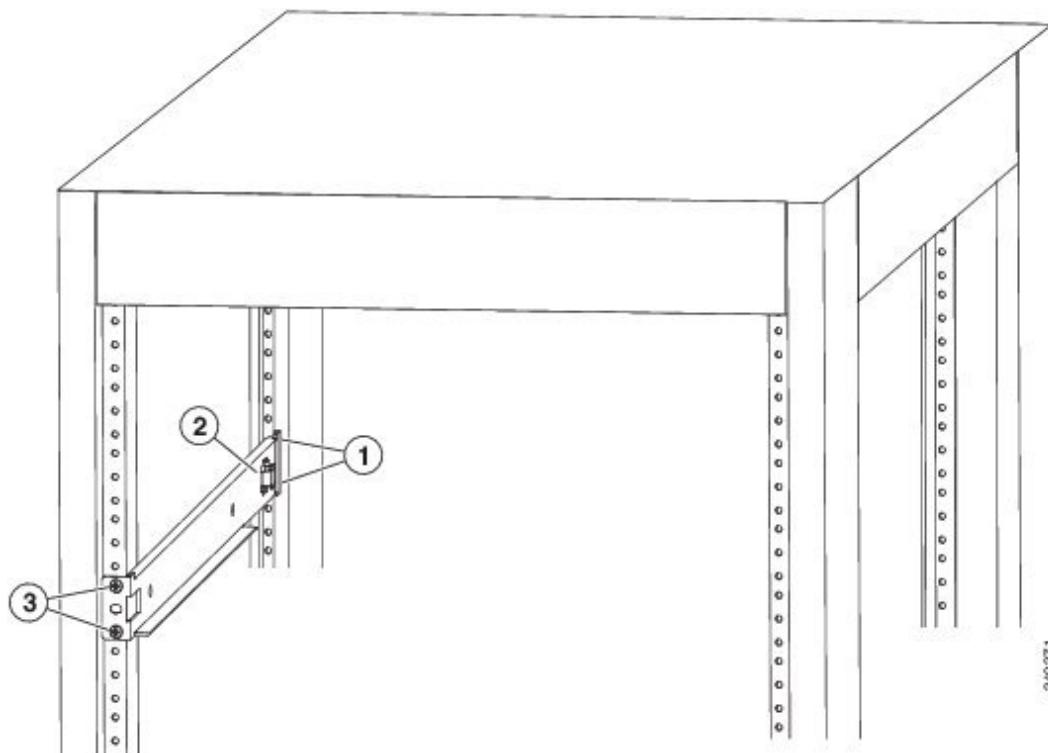
- If the fan modules have burgundy (port-side intake modules), position the bottom support rails so that the chassis stop is positioned by the hot aisle.

Step 2 Separate the two sliders that make up one bottom-support rail and position the half with the chassis stop by the appropriate aisle for the fan and power supply modules. Leave at least 1 rack unit open above the bottom-support rails so that you can easily install the chassis.

Step 3 Use two customer-supplied screws (typically M6 x 10 mm screws) to attach the bottom-support rail half to the vertical mounting rails on the rack post. Tighten each screw to the appropriate torque setting for the screws (for M6 x 10 mm screws, use 40 in. lbs [4.5 N·m] of torque).

Step 4 Slide the other half of the bottom-support rail onto the attached half of the rail set and use two customer supplied screws (typically M6 x 10 mm screws) to secure that portion to the vertical mounting rails on the rack. Tighten each screw to the appropriate torque setting for the screws (for M6 x 10 mm screws, use 40 in. lbs [4.5 N·m] of torque).

Figure 1: Positioning an Expanding Bottom-Support Rail Set



1	2 screws holding one end of the bottom-support bracket to the rear of the rack	3	2 screws holding the front end of the bottom-support bracket to the front side of the rack
2	Chassis stop on the expanding bottom-support bracket		

Step 5 Repeat Steps 2 and 3 to attach the other expanding bottom-support rails to the other side of the rack.

Note

Check the two installed bottom support rails to be sure that both have their chassis stops by the same aisle (either both by the hot aisle or both by the cold aisle) and that both rails are level and level with each other. If they are not level, adjust the higher rail down to the level of the lower rail.

What to do next

You are ready to install two front-mount brackets on the chassis.

Attaching Front-Mount Brackets to the Chassis

Attach a right-angled bracket to each side of the chassis. This bracket holds the chassis in place on a four-post rack.

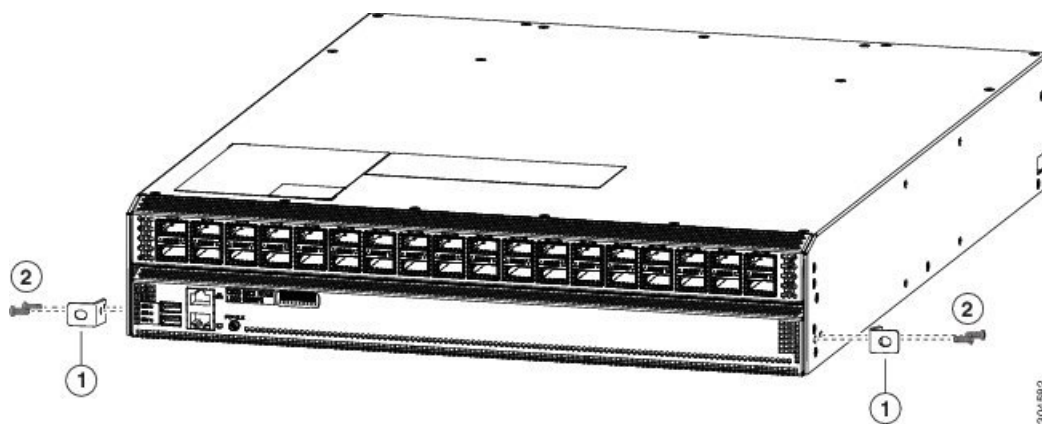
Before you begin

- You need these tools and equipment:
 - Manual Phillips-head torque screwdriver
 - Front-mount brackets (2) and screws (4). Check the switch accessory kit.

Procedure

Step 1 Align the two holes in one side of one of two front-mount brackets to two holes on the left or right side of the chassis (see the figure).

Be sure that the other side of the bracket is facing toward the front (port end) of the chassis.



1	Front-mount bracket with two screw holes aligned to two screw holes in the chassis and one screw hole facing the front (port side) of the chassis.	2	Two M4 x 6 mm screws used to fasten the bracket to the chassis.
---	--	---	---

Step 2 Use two M4 x 6 mm screws to attach the bracket to the chassis. Tighten each screw to 11 to 15 in-lb (1.2 to 1.7 Nm).

Step 3 Repeat Steps 1 and 2 to attach the second center-mount bracket to the other side of the chassis.

What to do next

You are ready to mount the chassis to the four-post rack.

Installing the Chassis in a Four-Post Rack

Slide the chassis onto the bottom-support rails so that the power supply end locks onto the chassis stops at the end of the rails and so that the front-mount brackets on the chassis come into contact with the front-mount rails on the rack.



Warning **Statement 1074**—Comply with Local and National Electrical Codes

To reduce risk of electric shock or fire, installation of the equipment must comply with local and national electrical codes.



Warning **Statement 1032**—Lifting the Chassis

To prevent personal injury or damage to the chassis, never attempt to lift or tilt the chassis using the handles on modules (such as power supplies, fans, or cards); these types of handles are not designed to support the weight of the unit.

Before you begin

- Make sure that the four-post rack is properly installed and secured to the concrete subfloor.
- Make sure that the bottom-support rails are installed so that the fan modules will be in the appropriate aisle as follows:
 - Burgundy (port-side intake airflow) fan modules are positioned in a hot aisle (the chassis stop on the bottom-support rails is positioned by the hot aisle).
 - Blue colored (port-side exhaust airflow) fan modules are positioned in a cold aisle (the chassis stop on the bottom-support rails is positioned by the cold aisle).
- Make sure that two front-mount brackets are securely fastened to the sides of the chassis at the port end.
- Make sure that you have two customer-supplied rack-mount screws (M6 x 10 mm or appropriate screw for the vertical mounting rails on the rack).

**Warning****Statement 1006**—Chassis Warning for Rack-Mounting and Servicing

To prevent bodily injury when mounting or servicing this unit in a rack, you must take special precautions to ensure that the system remains stable. The following guidelines are provided to ensure your safety:

- This unit should be mounted at the bottom of the rack if it is the only unit in the rack.
- When mounting this unit in a partially filled rack, load the rack from the bottom to the top with the heaviest component at the bottom of the rack.
- If the rack is provided with stabilizing devices, install the stabilizers before mounting or servicing the unit in the rack.

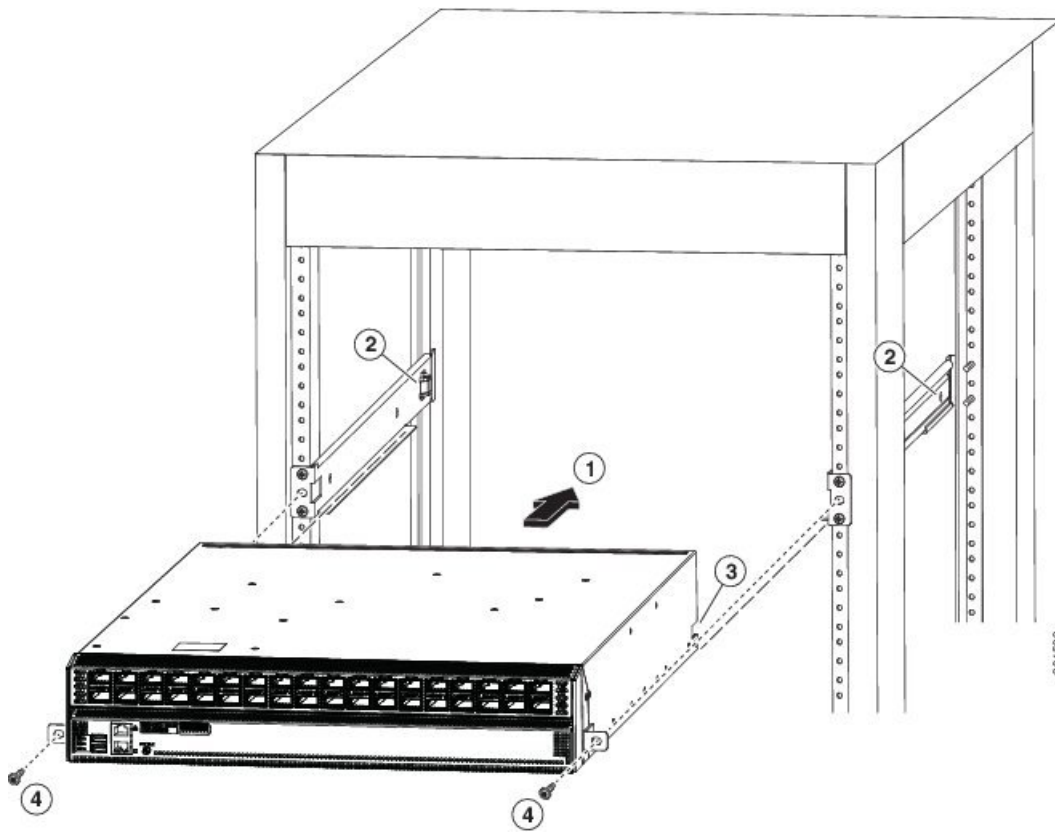
Procedure**Step 1**

Slide the power supply end of the chassis onto the bottom-support rails that are installed on the rack.

Be sure that the sides of the chassis by the power supplies clip into the chassis stops on the bottom-support rails and the front-mount brackets come in contact with the rack. See the figure.

Note

If the bottom-support rails are extended a long distance, they can bend outwards slightly when you install the chassis and the chassis stops at the far end of the rails might not fit into the end of the chassis. If this happens, press the side rails toward the sides of the chassis so that the chassis stops can go inside the chassis and hold it in place on the rack.



1	Slide the power-supply end of the chassis onto the bottom-support rails so that the chassis locks onto the chassis stops at the end of the rails.	3	Receiving hole on each side of the chassis for the chassis stops on the bottom-support rails.
2	Chassis stops for holding the chassis (positioned by the aisle required for the fan and power supply modules).	4	Customer-supplied rack-mount screw (M6 x 10 mm screw or other screw appropriate for the rack) used to secure each side of the chassis to the rack.

Step 2

Use a customer-supplied rack-mount screw (an M6 x 10 mm screw or other appropriate screw for the rack) to attach each of the two mounting brackets on the chassis to the rack and tighten each screw to the appropriate torque setting for the screw (for M6 x 10 mm screws, use 40 in-lbs [4.5 N·m] of torque).

Grounding the Chassis

The switch chassis is automatically grounded when you properly install the switch in a grounded rack with metal-to-metal connections between the switch and rack.



Note Provide an electrical conducting path between the product chassis and the metal surface of the enclosure or rack in which it is mounted or to a grounding conductor. To ensure electrical continuity, use thread-forming type mounting screws that remove any paint or non-conductive coatings and establish a metal-to-metal contact. Remove any paint or other non-conductive coatings on the surfaces between the mounting hardware and the enclosure or rack. Clean the surfaces and apply an antioxidant before installation.

Ground the rack if using LVDC power supplies. If using AC or HVDC power supplies, the power cord for the AC power supplies provides grounding for the chassis. For supplemental grounding or bonding, attach a customer-supplied grounding cable to the chassis ground pad.

Ground the chassis. If you are using a 2-post rack, attach a customer-supplied grounding cable. Attach the cable to the chassis grounding pad and the facility ground. If you are using a 4-post rack, ensure that your chassis is grounded through the rack mount system or the power cable (AC or HVDC).



Warning **Statement 1024**—Ground Conductor

This equipment must be grounded. To reduce the risk of electric shock, never defeat the ground conductor or operate the equipment in the absence of a suitably installed ground conductor. Contact the appropriate electrical inspection authority or an electrician if you are uncertain that suitable grounding is available.



Warning **Statement 1046**—Installing or Replacing the Unit

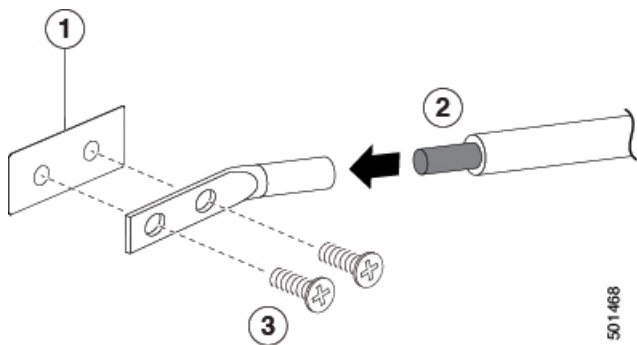
To reduce risk of electric shock, when installing or replacing the unit, the ground connection must always be made first and disconnected last.

Before you begin

Before you can ground the chassis, verify the earth ground contact has a solid connection to the data center building.

Procedure

-
- Step 1** Use a wire-stripping tool to remove approximately 0.75 inch (19 mm) of the covering from the end of the grounding wire. We recommend 6-AWG wire for the U.S. installations.
- Step 2** Insert the stripped end of the grounding wire into the open end of the grounding lug. Use a crimping tool to crimp the lug to the wire. See the figure. Verify that the ground wire is securely attached to the grounding lug by attempting to pull the wire out of the crimped lug (tug test).



1	Chassis grounding pad	3	2 M4 screws are used to secure the grounding lug to the chassis
2	Grounding cable, with 0.75 in. (19 mm) of insulation that is stripped from one end, which is inserted into the grounding lug and crimped in place		

- Step 3** Secure the grounding lug to the chassis grounding pad with two M4 screws, see figure 1. Tighten the screws to 11 to 15 in-lb (1.24 to 1.69 N m) of torque.
- Step 4** Prepare the other end of the grounding wire and connect it to the facility ground.

Start the switch

Start the switch by connecting it to its dedicated power source. If you need $n+n$ redundancy, connect each power supply in a switch to a different power source.



Note This equipment is designed to boot up in less than 30 minutes, dependent on its neighboring devices being fully up and running.

Before you begin

- The switch must be installed and secured to a rack or cabinet.
- The switch must be adequately grounded.
- The rack must be close enough to the dedicated power source so that you can connect the switch to the power source by using the designated power cables.
- You have the designated power cables for the power supplies that you are connecting to the dedicated power sources.



Note Depending on the outlet receptacle on your AC power distribution unit, you might need an optional jumper power cord to connect the switch to your outlet receptacle.

- The switch is not connected to the network (this includes any management or interface connections).
- The fan and power supply modules are fully secured in their chassis slots.

Procedure

- Step 1** For any AC power supply, do the following:
- a) Using the recommended AC power cable for your country or region, connect one end to the AC power supply.
 - b) Connect the other end of the power cable to the AC power source.
- Step 2** For any DC power supply, do the following:
- a) Turn off the circuit breaker for the power source to avoid an electrical shock hazard.
 - b) Verify that the power cable wires from the power source are connected to a connector block.
 - c) Insert the connector block into the receptacle on the power supply. Make sure that the connector block clicks when fully inserted in the receptacle and does not pull out.
 - d) If there is a safety cover for the terminals, place and secure it over the terminals to avoid an electrical shock hazard.
 - e) Turn on the power at the circuit breaker for the DC power source.
- Step 3** Verify that the power supply LED is on and green.
- Step 4** Listen for the fans; they should begin operating when the power supply is powered.
- Step 5** After the switch boots, verify that these LEDs are lit:
- On the fan modules, the Status (STA or STS) LED is green.
If a fan module Status LED is not green, try reinstalling the fan module.
 - After initialization, the switch chassis Status (labeled as STA or STS) LED is green.
- Step 6** Verify that the system software has booted and the switch has initialized without error messages.
- A setup utility automatically launches the first time that you access the switch and guides you through the basic configuration. For instructions on how to configure the switch and check module connectivity, see the appropriate [Cisco Nexus 9000 Series Configuration Guides](#).
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CHAPTER 4

Connecting the Switch to the ACI Fabric

- [ACI Fabric Topology, on page 31](#)
- [Preparing to Connect to Other Devices, on page 32](#)
- [Connecting Leaf Switches to APICs, on page 33](#)
- [Connecting Leaf Switches to Spine Switches, on page 35](#)
- [Installing a Gigabit Ethernet module \(GEM\), on page 36](#)
- [Replacing an Existing Switch in the ACI Fabric, on page 37](#)
- [Setting Up an Optional Console Interface, on page 37](#)
- [Setting Up an Optional Management Connection, on page 37](#)
- [Maintaining Transceivers and Optical Cables, on page 38](#)

ACI Fabric Topology

The ACI fabric topology includes these major components:

- Application Centric Infrastructure Controller (APIC) appliance (cluster of APICs)
- Leaf switches (for switch compatibility, please see the [data sheets](#))
- Spine switches (for switch compatibility, please see the [data sheets](#))

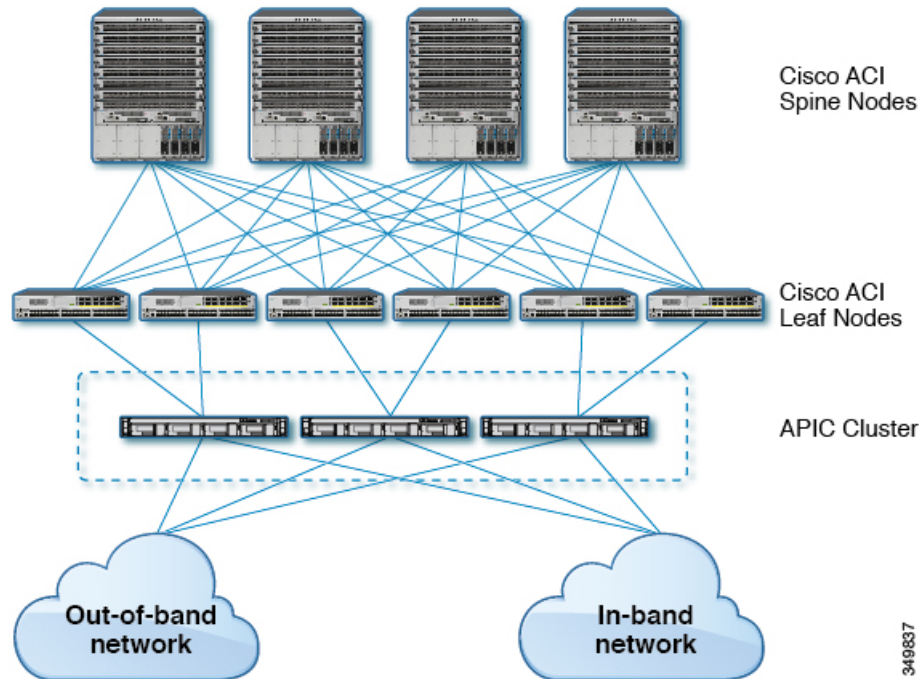
For additional information, please see the [Cisco APIC Installation, Upgrade, and Downgrade Guide](#).



Note To prevent sub-optimal forwarding between endpoints, connect every leaf switch in the fabric to every spine switch in the same fabric.

As shown in the figure, each APIC is connected to one or two leaf switches and each leaf switch should be connected to every spine switch in the same fabric.

Figure 2: Connections Between APIC Clusters, Leaf Nodes, and Spine Nodes



349837

Preparing to Connect to Other Devices

When preparing to connect the fabric devices, consider these items for each type of interface, and gather all of the required equipment before making the connections:

- Cabling type required for each interface type
- Distance limitations for each signal type
- Additional interface equipment required



Note When running power and data cables in overhead or subfloor cable trays, we strongly recommend that you locate power cables and other potential noise sources as far away as practical from network cabling that terminates on Cisco equipment. In situations where long parallel cable runs cannot be separated by at least 3.3 feet (1 meter), we recommend that you shield any potential noise sources by housing them in a grounded metallic conduit.

The optical transceivers that are not already assembled to their cables come separate from their cables. To prevent these transceivers and their cables from being damaged, we recommend that you keep the transceivers disconnected from their cables when installing them in ports and then insert the optical cable into the transceiver. When removing transceivers from ports, remove their cables before removing the transceivers.

For more information about removing and installing the transceivers, see the [Cisco Nexus 9408 ACI-Mode Switch Hardware Installation Guide](#), "Optic Transceiver Removal Using the Optics Extraction Tool".

To maximize the effectiveness and life of your transceivers and optical cables:

- Wear an ESD-preventative wrist strap that is connected to an earth ground whenever handling transceivers. The switch is typically grounded when you install transceivers and provides an ESD port to which you can connect your wrist strap. If you cannot find an ESD port, connect the wrist strap to an earth ground (such as the grounding connection for the chassis).
- Do not remove or insert a transceiver more often than necessary. Repeated removals and insertions can shorten its useful life.
- Keep the transceivers and fiber-optic cables clean and dust free to maintain high signal accuracy and to prevent damage to the connectors. Attenuation (loss of light) increases with contamination and should be kept below 0.35 dB.
 - Clean these parts before installing them to prevent dust from scratching the fiber-optic cable ends.
 - Clean the connectors regularly; the required frequency of cleaning depends upon the environment. In addition, clean connectors if they are exposed to dust or accidentally touched. Both wet and dry cleaning techniques can be effective; refer to your site's fiber-optic connection cleaning procedures.
 - Do not touch the ends of connectors. Touching the ends can leave fingerprints and cause other contamination.
- Inspect routinely for dust and damage. If you suspect damage, clean and then inspect fiber ends under a microscope to determine if damage has occurred.

Connecting Leaf Switches to APICs

Downlink one or two (recommended for redundancy) Cisco Nexus 9000 platform ACI-mode leaf switches to each Application Policy Infrastructure Controller (APIC) in your ACI fabric. The type of virtual interface card (VIC) installed on the APIC determines the types of interface cables that you use to connect the leaf switches to the APICs.

- The **VIC 1225T** module supports copper connectors, copper cables, and switches with copper downlink ports (such as: Cisco Nexus 93108TC-FX switch).
- The **VIC 1225** module supports optical transceivers, optical cables, and switches with optical downlink ports (such as: Cisco Nexus 93600CD-GX switch).
- The **VIC 1455** module supports optical transceivers, optical cables, and switches with optical downlink ports (such as: Cisco Nexus 93600CD-GX switch).



Note You cannot use breakout ports for Cisco APIC connectivity.

Before you begin

The APIC and leaf switches in the fabric must be fully installed in their racks and grounded.

Procedure

Step 1

Connect an interface cable to one of the two to four ports on the virtual interface card (VIC) installed on the APIC. If the cable is not already assembled to its transceiver, insert the transceiver into the VIC port. Connect the optical interface cable to the transceiver.

- For a **VIC 1225T** 10GBASE-T copper module, use 10GBASE-T cables with RJ-45 connectors.
- For a **VIC 1225** optical module, use one of these sets of transceivers and cables:
 - Cisco 10GBASE-LR transceivers (SFP-10G-LR) supporting a link length of up to 6.1 miles (10 km)
 - Cisco 10GBASE-SR transceivers (SFP-10G-SR) supporting these link lengths:
 - Using 2000 MHz MMF (OM3) for up to 984 feet (300 m)
 - Using 4700 MHz MMF (OM4) for up to 1312 feet (400 m)
 - Cisco SFP+ Active Optical Cables (SFP-10G-AOCxM [where x=1, 3, 5, 7 for lengths in meters])
 - Cisco SFP+ Twinax Cables (SFP-H10GB-CUxM [where x=7 for lengths in meters])
- For a **VIC 1455** SFP28 module, 10-Gigabit only, use one of these sets of transceivers and cables:
 - Cisco 10GBASE-LR transceivers (SFP-10G-LR) supporting a link length of up to 6.1 miles (10 km)
 - Cisco 10GBASE-SR transceivers (SFP-10G-SR) supporting these link lengths:
 - Using 2000 MHz MMF (OM3) for up to 984 feet (300 m)
 - Using 4700 MHz MMF (OM4) for up to 1312 feet (400 m)
 - Cisco SFP+ Active Optical Cables (SFP-10G-AOCxM [where x=1, 2, 3, 5, 7, or 10 for lengths in meters])
 - Cisco SFP+ Twinax Cables (SFP-H10GB-CUxM [where x=1, 2, 3, 5, 7, or 10 for lengths in meters])

Note

The VIC 1455 has 4 ports: port-1, port-2, port-3, and port-4 from left to right.

- All ports must have the same speed, either 10-Gigabit or 25-Gigabit.
- Port-1 and port-2 is one pair, corresponding to eth2-1 on APIC and port-3 and port-4 is another pair, corresponding to eth2-2 on APIC. Only one connection is allowed for each pair. For example, connect one cable to either port-1 or port-2. Connect another cable to either port-3 or port-4. **Do not connect two cables on any pair.**

To determine which transceivers, adapters, and cables support this switch, see the [Cisco Transceiver Modules Compatibility Information](#) document.

To see the transceiver specifications and installation information, see [Transceiver Module Installation Guides](#).

Step 2

Connect the other end of the interface cable to a downlink port on a leaf switch.

- For a Cisco 10GBASE-LR or -SR transceiver and cable, insert the transceiver into a downlink optical port on a leaf switch before connecting the cable to the transceiver.

- For Cisco SFP+ Active Optical Cables, insert the transceiver on the cable into a downlink optical port on a leaf switch.
- For a 10GBASE-T copper cable, insert the RJ-45 connector on the cable into a downlink BASE-T port on a leaf switch.

Note

To determine which transceivers, adapters, and cables support this switch, see the [Cisco Transceiver Modules Compatibility Information](#) document.

Connecting Leaf Switches to Spine Switches

For optimal forwarding between endpoints, connect each leaf switch to every spine switch in the same ACI fabric.

To determine which transceivers, adapters, and cables support this switch, see the [Cisco Transceiver Modules Compatibility Information](#) document.

To see the transceiver specifications and installation information, see [Transceiver Module Installation Guides](#).

**Warning Statement 1055—Class 1/1M Laser**

Invisible laser radiation is present. Do not expose to users of telescopic optics. This applies to Class 1/1M laser products.

**Warning Statement 1056—Unterminated Fiber Cable**

Invisible laser radiation may be emitted from the end of the unterminated fiber cable or connector. Do not view directly with optical instruments. Viewing the laser output with certain optical instruments, for example, eye loupes, magnifiers, and microscopes, within a distance of 100 mm, may pose an eye hazard.

**Warning Statement 1255—Laser Compliance Statement**

Pluggable optical modules comply with IEC 60825-1 Ed. 3 and 21 CFR 1040.10 and 1040.11 with or without exception for conformance with IEC 60825-1 Ed. 3 as described in Laser Notice No. 56, dated May 8, 2019.

Before you begin

- The leaf and spine switches in the fabric (such as: N9K-C9364C, N9K-C9332C, N9K-C9316D-GX, and N9K-C9332D-GX2B) must be fully installed in their racks and grounded.

- If there are modular switches in the fabric, their ACI-mode line cards must already be installed. The line cards can be of these types:
 - 32-port 100-Gigabit (such as: N9K-X9732C-EX)
 - 36-port 100-Gigabit (such as: N9K-X9736C-FX)
 - N9K-X9400-22L linecard expansion module (LEM)
 - N9K-X9400-16W LEM
 - N9K-X9400-8D LEM

**Note**

- You cannot include NX-OS line cards in the same chassis when running in ACI mode.
- Multiple uplinks from a leaf switch to a spine switch is supported. A symmetrical topology is recommended so that all devices have equal access to resources.

Procedure

-
- Step 1** For the transceivers with removable cables, separate the transceivers from their interface cables.
 - Step 2** Insert the appropriate transceiver into an active uplink port on the leaf switch.
 - Step 3** Insert the same type of transceiver in the spine switch port on the line card.
 - Step 4** For transceivers with removable cables, insert the interface cable into the open end of each of those transceivers.
 - Step 5** Repeat Steps 1 through 4 for each spine switch in the ACI fabric.
The leaf switch is connected to each spine switch in the ACI fabric.
 - Step 6** Repeat Steps 1 through 5 for each leaf switch in the ACI fabric.
Each leaf switch in the ACI fabric is connected to each spine switch in the network.
-

The fabric automatically implements Equal Cost Multi-Pathing (ECMP) and enables all links. You do not need to configure the links.

Installing a Gigabit Ethernet module (GEM)

Procedure

-
- Step 1** Clear the switch's current configuration. Use the `setup-clean-config` command.
 - Step 2** Power off the switch; disconnect the power.
 - Step 3** Replace the current GEM card with the new GEM card.
 - Step 4** Power on the switch.
-

Replacing an Existing Switch in the ACI Fabric

To replace an existing switch in the ACI fabric, see "Upgrading the Switch Hardware" in the [Cisco APIC Installation and ACI Upgrade and Downgrade Guide](#).

Setting Up an Optional Console Interface

Optionally set up a console interface for performing the initial configuration of the switch. Use the interface cable provided in the accessory kit to connect the switch to your console device. Connect the console port on the switch to a modem. If you do not connect it to a modem, make the connection either before powering up the switch or after completing the boot process for the switch.

Before you begin

The console device must support VT100 terminal emulations and asynchronous transmissions.

Procedure

-
- | | |
|---------------|---|
| Step 1 | Configure the terminal emulator program to match each of these default port characteristics: <ul style="list-style-type: none">• 9600 baud• 8 data bits• 1 stop bit• No parity |
| Step 2 | Insert the RJ-45 connector on the interface cable found in the accessory kit into the RS-232 port on the switch. Insert the DB-9 connector on the other end of the cable to the serial port on the console device. |
-

What to do next

Perform the initial configuration for the switch (see the *Cisco ACI Getting Started Guide*).

Setting Up an Optional Management Connection

Optionally set up an out-of-band management connection for monitoring and troubleshooting purposes. Depending on your switch, connect either the RJ-45 management port or the SFP management port on the switch to an external hub, switch, or router.

Before you begin

To prevent an IP address conflict, complete the initial configuration for the switch and establish an IP address before you create the management connection.

Procedure

- Step 1** Connect the interface cable to a management port on the switch.
- Step 2** Connect the other end of the cable to an external hub, switch, or router.
-

Maintaining Transceivers and Optical Cables

Keep transceivers and fiber-optic cables clean and dust free to maintain high signal accuracy and prevent damage to the connectors. Contamination increases attenuation (loss of light) and should be below 0.35 dB.

Consider these maintenance guidelines:

- Transceivers are static sensitive. To prevent ESD damage, wear an ESD-preventative wrist strap that is connected to the grounded chassis.
- Do not remove and insert a transceiver more often than is necessary. Repeated removals and insertions can shorten its useful life.
- Keep all optical connections covered when not in use. Clean them before using to prevent dust from scratching the fiber-optic cable ends.
- Do not touch the ends of connectors. Touching the ends can leave fingerprints and cause other contamination.
- Clean the connectors regularly; the required frequency of cleaning depends upon the environment. In addition, clean connectors if they are exposed to dust or accidentally touched. Both wet and dry cleaning techniques can be effective; refer to the fiber-optic connection cleaning procedures for your site.
- Inspect routinely for dust and damage. If you suspect damage, clean and then inspect fiber ends under a microscope to determine if damage has occurred.



Note When you need to remove a fiber-optic transceiver, first remove the fiber-optic cable from the transceiver **before** you remove the transceiver from the port.



CHAPTER 5

Replacing Modules

- [Replacing a Fan Module During Operations, on page 39](#)
- [Replacing a Power Supply Module, on page 41](#)

Replacing a Fan Module During Operations

Before you begin

Before you can replace a fan module, ensure that these conditions exist:

- There is another functioning fan module in the chassis. In order to replace a fan module during operations, there must be another fan module circulating air in the chassis at all times.
- The replacement fan module must have the same airflow direction as the other modules in the chassis.

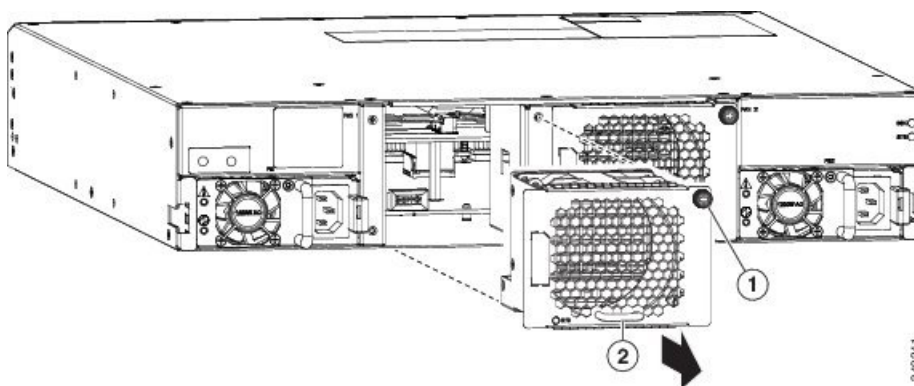
If you must replace the fan module during operations and the above conditions are not met, leave the fan module that you need to replace in the chassis to preserve the designed airflow until you have the required module.

Procedure

Step 1

Remove the fan module that you need to replace.

- a) Unscrew the captive screw that secures the fan module to the chassis (see the figure).



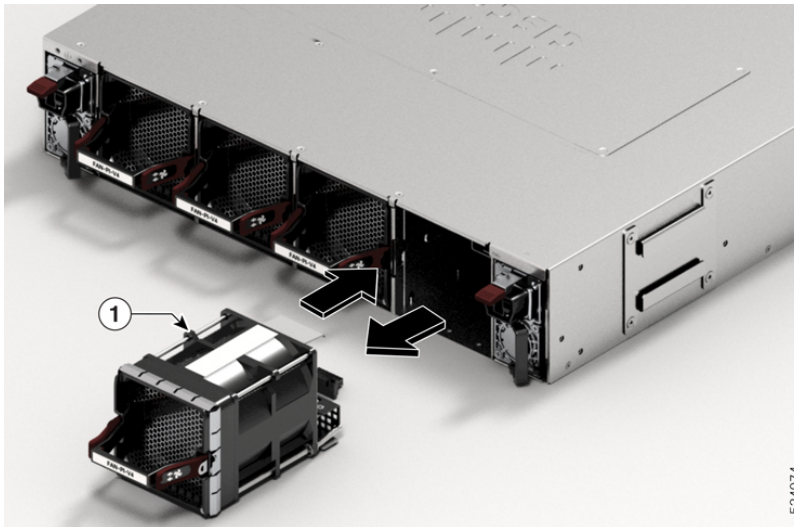
1	Loosen the captive screw so that it is no longer in contact with the chassis.	2	Holding the handle on the fan module, slide it out of the chassis.
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- b) Pull the fan module handle to slide the module out of the chassis.

Caution

To prevent electrostatic discharge (ESD) damage to the module electronics, do not touch the electrical connectors on the back side of the module. Also, to prevent any damage to the electrical connectors, prevent them touching anything that can bend or break them.

- c) Press two latches on the fan module and grasp the handle of fan module.



- d) Place the removed module on an antistatic surface or in an antistatic bag. If possible, repack the module in its packing materials for safe shipping or storage.
- e) As you press the latches, pull the fan module fully out of the chassis.
- f) Place the removed module on an antistatic surface or in an antistatic bag. If possible, repack the module in its packing materials for safe shipping or storage.

1	Latched Fan Module
---	--------------------

Step 2

Use these steps to replace the missing fan module within two minutes to avoid a shutdown.

- a) Remove the replacement fan module from its packing materials and place it on an antistatic surface.

Hold the module by its handle and do not touch the electrical connectors on its backside. Also, to protect the electrical connectors, avoid letting them come in contact with anything other than the electrical connectors inside the chassis.

- b) Verify that you have the right fan module for the chassis. The correct fan module has one of these part numbers:
- N9K-C9300-FAN3-B (port-side exhaust airflow direction and a blue stripe)
 - N9K-C9300-FAN3 (port-side intake airflow direction and a burgundy stripe)

Note

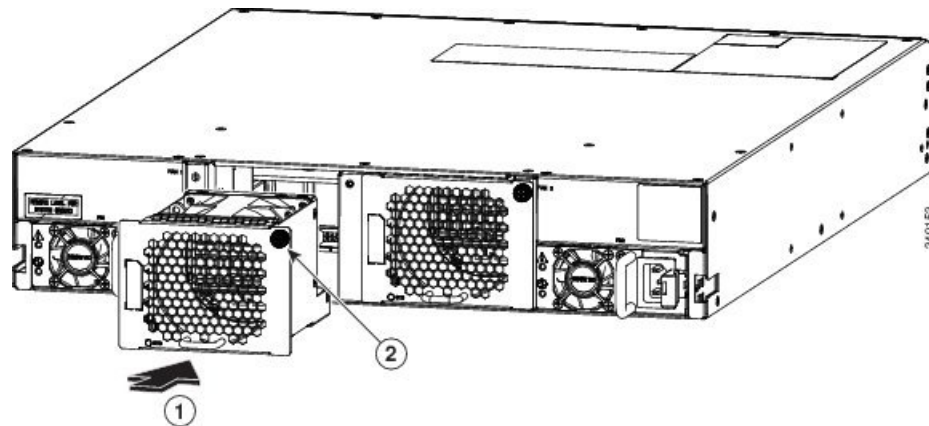
Be sure that the airflow direction of the new fan module matches the airflow direction of the other fan and AC power supply modules already installed in the chassis. Port-side intake airflow is indicated with burgundy or green coloring,

and port-side exhaust airflow is indicated with blue or gray coloring. Power supplies with dual-direction airflow (airflow direction is set by the fan modules) have white coloring.

- c) Position the fan module in front of the open fan slot (be sure that the backside of the module with the electrical connectors is positioned to enter the slot first) and slide the module all the way into the chassis until its front side comes in contact with the chassis. For the last 0.2 inches (0.5 cm), carefully mount the module onto the chassis connectors by pushing more firmly, but do not force the module if it does not move further (excessive force can damage the connectors).

Note

If you are not able to push the module all the way into the slot, carefully slide the module out of the slot and check its electrical connectors for damage. If damaged, contact Cisco Technical Assistance for help. If undamaged, repeat this step to reinstall the module.



1	Holding the handle on the fan module, slide its connector end into the fan slot until the front of the fan module is touching the chassis.	2	Screw in the captive screw to attach the module to the chassis.
---	--	---	---

- d) Verify that the STS LED turns on and becomes green.

If the STS LED does not turn on, slide the module out of the chassis, and visually check the electrical connectors on the back side of the chassis for damage. If damaged, contact Cisco Technical Assistance for help. If undamaged, repeat the previous step to reinstall the module.

- e) Secure the fan module to the chassis by screwing in the captive screw to the chassis. Tighten the screw to 5 to 7 in-lb (0.56 to 0.79 N·m).

Replacing a Power Supply Module

The switch requires two power supplies for redundancy. With one power supply providing the necessary power for operations, replace the other power supply during operations as long as the new power supply has the same airflow direction as the other modules in the chassis.

Replace a power supply with another supported power supply that has the same power source type as the other installed power supply. Additionally, the airflow direction of the power supply must match or conform

to the airflow direction of the installed fan modules. For the airflow direction used by the switch, see the coloring of the fan modules.

Removing an AC Power Supply

To remove an AC power supply, disconnect the power cable and remove the module from the chassis.

Before you begin

- To replace a power supply during operations, there must be a functioning power supply providing power to the switch while you replace the other power supply. If there is only one power supply installed in the chassis and you need to replace it, install the new power supply in the open slot and power it up before removing the original power supply.
- Ensure that the chassis is grounded. For grounding instructions, see [Grounding the Chassis, on page 27](#).

Procedure

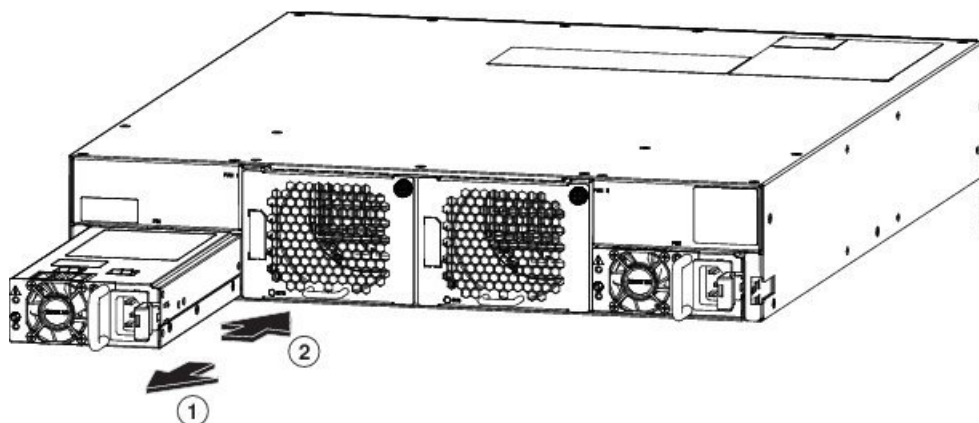
Step 1 Disconnect the power cord from the power receptacle on the power supply to be removed. Verify that the LED turns off.

Note

The LED might be on and amber colored. This indicates that the input power has been disconnected.

Step 2 Remove the power supply from the chassis by pushing and holding its thumb latch to the left and pulling the power supply part way out of the chassis (see the figure).

Figure 3: Replacing a Power Supply from the Chassis



1	Push thumb latch to the left and slide out to remove.	2	Slide in until the latch clicks to install
---	---	---	--

Step 3 Place your other hand under the power supply to support it while you slide it out of the chassis. Either place the power supply on an antistatic surface or pack it in its packing materials.

What to do next

Install the replacement power supply.

Removing a DC Power Supply

You can remove one power supply while the other one provides power to the switch.

To disconnect the power supply from its power cables, shut off the power from the power source and disconnect the connector for the power cables.

Procedure

-
- | | |
|---------------|---|
| Step 1 | Turn off the circuit breaker for the power feed to the power supply that you are replacing.
Verify that the LEDs turn off on the power supply that you are removing. |
| Step 2 | Remove the power cable from the power supply by pressing the latch on the Cisco-supplied power cord to allow removal from the power supply. <ul style="list-style-type: none">• Use the supplied cable to connect the power supply to external power. |
| Step 3 | Grasp the power supply handle while pressing the release latch towards the power supply handle. |
| Step 4 | Pull the power supply out of the bay. |
-

What to do next

Install a DC power supply in the open slot.

Installing an AC Power Supply

You can replace one power supply while the other one provides power to the switch.

Before you begin

- The power supply that you are installing must be capable of using the same airflow direction as the fan trays installed in the same switch. It must use the same type of power source as the other power supply installed in the same switch. (Do not mix AC and DC power supplies in the same switch.)



Note	DC power supplies with green coloring have the same port-side intake airflow direction as the power supplies with red coloring, and DC power supplies with gray coloring have the same port-side exhaust airflow direction as the power supplies with blue coloring. If the power supply that you are replacing has a different color handle than the replacement power supply, verify that it has or will have the same airflow direction as the other modules in the switch.
-------------	--

- An AC power source must be within reach of the power cable that will be used with the replacement power supply. If you are using $n+n$ power redundancy, there must be a separate power source for each power supply installed in the chassis. Otherwise, only one power source is required.
- There must be an earth-ground connection to the chassis that you are installing the replacement module. Typically, the chassis is grounded by its metal-to-metal connection with a grounded rack. If you need to ground the chassis, see [Grounding the Chassis, on page 27](#).

Procedure

Step 1 Holding the replacement power supply with one hand underneath the module and the other hand holding the handle, turn the power supply so that its release latch is on the right side. Align the back end of the power supply (the end with the electrical connections) to the open power supply slot. Carefully slide the power supply all the way into the slot until it clicks into place.

Note

If the power supply does not fit into the open slot, turn the module over, before sliding it carefully into the open slot.

Step 2 Test the installation by trying to pull the power supply out of the slot without using the release latch. If the power supply does not move out of place, it is secured in the slot. If the power supply moves, carefully press it all the way into the slot until it clicks into place.

Step 3 Attach the power cable to the electrical inlet on the front of the power supply.

Step 4 Verify that the other end of the power cable is attached to the appropriate power source for the power supply.

Step 5 Verify that the power supply is operational by making sure that the power supply  LED is green.

Installing a DC Power Supply

This topic is for installing the 48-V DC power supply into the switch chassis.

You can replace one power supply while the other one provides power to the switch.

Before you begin

- The circuit breaker for the DC power source for the power supply must be turned off.
- The power supply that you are installing must be capable of using the same airflow direction as the fan trays installed in the same switch.
- A DC power source must be within reach of the power cable that will be used with the replacement power supply. If you are using $n+n$ power redundancy, there must be a separate power source for each power supply installed in the chassis. Otherwise, only one power source is required.
- There must be an earth-ground connection to the chassis that you are installing the replacement module. Typically, the chassis is grounded by its metal-to-metal connection to a grounded rack. If you need to ground this chassis by another means, see [Grounding the Chassis, on page 27](#).

- All DC power supplies have reverse polarity protection. When you inadvertently connect the input power (+) to the DC PSU's – terminal and the input power – to the DC PSU's (+) terminal, the PSU will not be damaged and will operate fine after the input power feeds are correctly wired.

Procedure

- Step 1** Holding the replacement power supply with one hand underneath the module and the other hand holding the handle, turn the power supply so that its release latch is on the right side. Align the back end of the power supply (the end with the electrical connections) to the open power supply slot. Carefully slide the power supply all the way into the slot until it clicks into place.
- Note**
If the power supply does not fit into the open slot, turn the module over. Carefully slide it into the open slot.
- Step 2** Verify the rack is properly grounded, so that when the chassis is installed in the rack, it will be grounded.
- Step 3** Turn on the circuit breaker for the DC power source connected to the power supply.
- Step 4** Verify that the power supply is operational by making sure that the power supply  LED is green.

Wiring a 48 V DC Electrical Connector Block

You must connect the ground, negative, and positive DC power cables to a connector block in order to connect the power cables to a 48 V DC power supply.



Note The recommended wire gauge is 8 AWG. The minimum wire gauge is 10 AWG.



Warning **Statement 342**—Before Connecting to System Power Supply

High touch/leakage current – Permanently connected protective earth ground is essential before connecting to system power supply.



Warning **Statement 1024**—Ground Conductor

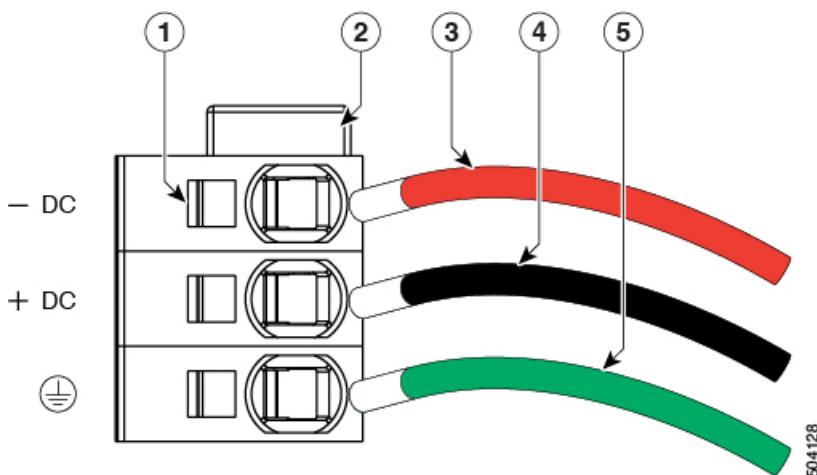
This equipment must be grounded. To reduce the risk of electric shock, never defeat the ground conductor or operate the equipment in the absence of a suitably installed ground conductor. Contact the appropriate electrical inspection authority or an electrician if you are uncertain that suitable grounding is available.

Before you begin

You must turn off the circuit breaker for the DC power cables that you are connecting to prevent electrocution.

Procedure

- Step 1** Verify that the circuit breaker for the power feed to the replacement power supply is turned off.
- Step 2** Remove the DC power connector block from the power supply by doing the following:
- Push the orange plastic button on the top of the connector block inward toward the power supply.
 - Pull the connector block out of the power supply.
- Step 3** Strip 0.6 inches (15 mm) of insulation off the DC wires that you are using.
- Step 4** Orient the connector as shown in the following figure with the orange plastic button on top.



1	Wire retainer lever	4	-48V (+DC) cable
2	Orange plastic button on top of the connector	5	Grounding cable (8 AWG recommended)
3	-48V (-DC) cable		

- Step 5** Use a small screwdriver to depress the spring-loaded wire retainer lever on the lower spring-cage wire connector. Insert your green (ground) wire into the aperture and then release the lever.
- Step 6** Use a small screwdriver to depress the spring-loaded wire retainer lever on the upper spring-cage wire connector. Insert your red (DC negative) wire into the aperture and then release the lever.
- Step 7** Use a small screwdriver to depress the spring-loaded wire retainer lever on the middle spring-cage wire connector. Insert your black (DC positive) wire into the aperture and then release the lever.
- Step 8** Insert the connector block back into the power supply. Make sure that your black (DC positive) wire aligns with the power supply label, "+ DC".
- Step 9** Verify that the other ends of the cables are attached to the DC power source and ground. You are then ready to turn on the DC power source.



APPENDIX **A**

Rack Specifications

- [Overview of Racks, on page 47](#)
- [General Requirements for Cabinets and Racks, on page 47](#)
- [Requirements Specific to Standard Open Racks, on page 48](#)
- [Requirements Specific to Perforated Cabinets, on page 48](#)
- [Cable Management Guidelines, on page 48](#)

Overview of Racks

Install the switch in these types of cabinets and racks, assuming an external ambient air temperature range of 0 to 104°F (0 to 40°C):

- Standard perforated cabinets
- Solid-walled cabinets with a roof fan tray (bottom to top cooling)
- Standard open racks



Note

- If you are using an enclosed cabinet, we recommend one of the thermally validated types, either standard perforated or solid-walled with a fan tray.
- We do not recommend using racks that have obstructions (such as power strips). The obstructions could impair access to field-replaceable units (FRUs).

General Requirements for Cabinets and Racks

The cabinet or rack must meet these requirements:

- Standard 19-inch (48.3 cm) (two- or four-post EIA cabinet or rack, with mounting rails that conform to English universal hole spacing per section 1 of ANSI/EIA-310-D-1992). For more information, see [Requirements Specific to Perforated Cabinets, on page 48](#).

The spacing between the posts of the rack must be (EIA-310-D-1992 19-inch rack compatible) wide enough to accommodate the width of the chassis.

- The minimum vertical rack space requirement per chassis is:
 - For a one RU (rack unit) switch, 1.75 inches (4.4 cm)
 - For a one and a half RU (rack unit) switch, 2.63 (6.68 cm)
 - For a two RU (rack unit) switch, 3.5 inches (8.8 cm)
 - For a three RU (rack unit) switch, 5.25 inches (13.3 cm)
- The width between the rack-mounting rails must be at least 17.75 inches (45.0 cm) if the rear of the device is not attached to the rack. For four-post EIA racks, this measurement is the distance between the two front rails.

Four-post EIA cabinets (perforated or solid-walled) must meet these requirements:

- The minimum spacing for the bend radius for fiber-optic cables should have the front-mounting rails of the cabinet offset from the front door by a minimum of 3 inches (7.6 cm).
- The distance between the outside face of the front mounting rail and the outside face of the back mounting rail should be 23.0 to 30.0 inches (58.4 to 76.2 cm) to allow for rear-bracket installation.

Requirements Specific to Standard Open Racks

If you are mounting the chassis in an open rack (no side panels or doors), ensure that the rack meets these requirements:

- The minimum vertical rack space per chassis must be equal to the rack unit (RU) of the chassis. One rack unit is equal to 1.75 inches (4.4 cm).
- The distance between the chassis air vents and any walls should be 2.5 inches (6.4 cm).

Requirements Specific to Perforated Cabinets

A perforated cabinet has perforations in its front and rear doors and side walls. Perforated cabinets must meet these requirements:

- The front and rear doors must have at least a 60 percent open area perforation pattern, with at least 15 square inches (96.8 square cm) of open area per rack unit of door height.
- The roof should be perforated with at least a 20 percent open area.
- The cabinet floor should be open or perforated to enhance cooling.

The Cisco R Series rack conforms to these requirements.

Cable Management Guidelines

To help with cable management, allow additional space in the rack above and below the chassis to make it easier to route all of the fiber optic or copper cables through the rack.



APPENDIX **B**

System Specifications

- [Environmental Specifications, on page 49](#)
- [Switch Dimensions, on page 49](#)
- [Switch and Module Weights and Quantities, on page 49](#)
- [Switch Power Input Requirements, on page 50](#)
- [Power Cable Specifications, on page 50](#)

Environmental Specifications

Environment		Specification
Temperature	Ambient operating temperature	32 to 104°F (0 to 40°C)
	Ambient nonoperating	–40 to 158°F (–40 to 70°C)
Relative humidity	Operating	5 to 90%
	Nonoperating	5 to 95%
Altitude	Operating	0 to 13,123 feet (0 to 4,000 meters)

Switch Dimensions

Switch	Width	Depth	Height
Cisco Nexus 9336PQ	17.5 inches (44.5 cm)	22.5 inches (57.1 cm)	3.5 inches (8.9 cm) (2 RU)

Switch and Module Weights and Quantities

Component	Weight per Unit	Quantity
Cisco Nexus 9336PQ Chassis (N9K-C9336PQ)	24.3 lb (11.0 kg)	1

Component	Weight per Unit	Quantity
Fan Module	—	2
– Port-side exhaust (blue) (N9K-C9300-FAN3-B)	2.4 lb (1.1 kg)	
– Port-side intake (burgundy) (N9K-C9300-FAN3)	2.4 lb (1.1 kg)	
Power Supply Module	—	2 (1 for operations and 1 for redundancy)
– 1200-W AC port-side exhaust (blue) (N9K-PAC-1200W-B)	2.64 lb (1.2 kg)	
– 1200-W AC port-side intake (burgundy) (N9K-PAC-1200W)	2.64 lb (1.2 kg)	
– 1200-W HVAC/HVDC dual-direction (white) (N9K-PUV-1200W)	2.42 lb (1.1 kg)	
– 930-W DC port-side exhaust (gray) (UCS-PSU-6332-DC)	2.42 lb (1.1 kg)	
– 930-W DC port-side intake (green) (UCSC-PSU-930WDC)	2.42 lb (1.1 kg)	

Switch Power Input Requirements

This table lists the typical amount of power that the switch consumes. It also lists the maximum amount of power that you must provision for the switch and power supply for peak conditions.



Note Some power supplies have capabilities that are greater than the maximum power requirements for a switch. To determine the power consumption characteristics for the switch, use the typical and maximum requirements that are listed here.



Note If you want to use optics that consume **5W or more power on all the 36 ports**, you must install 1100W power supplies.

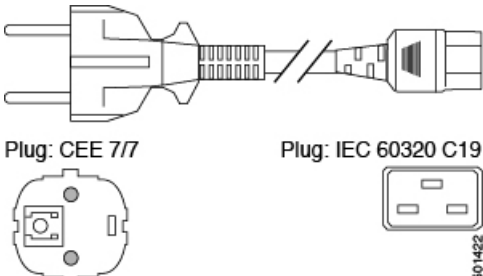
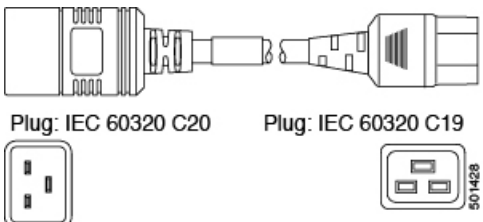
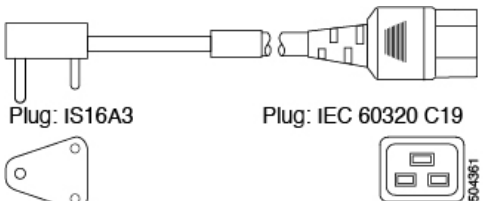
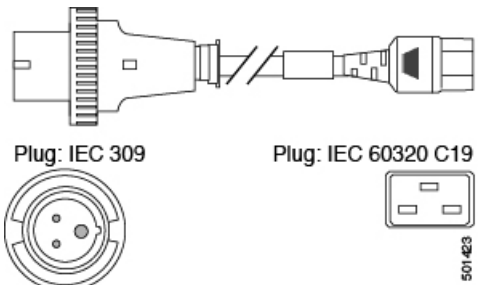
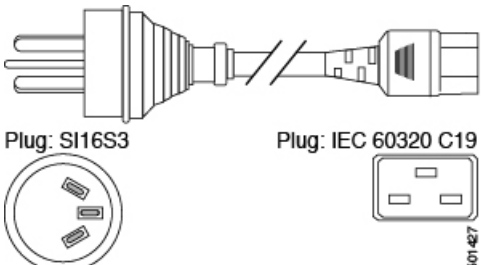
Switch	Typical Power Consumption (AC or DC)	Maximum Power Consumption (AC or DC)	Heat Dissipation Requirement
Cisco Nexus 9336PQ	400 W	660 W	2252.013 BTUs per hour

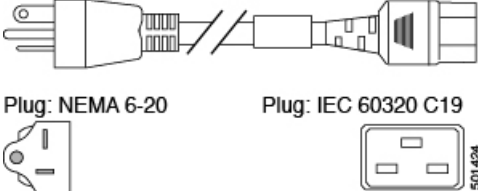
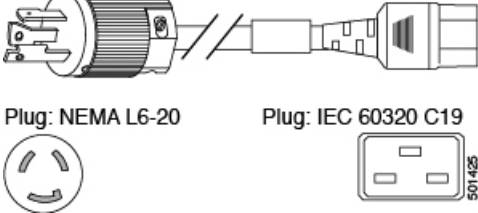
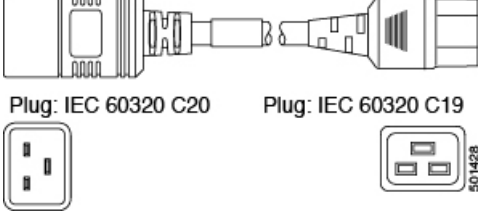
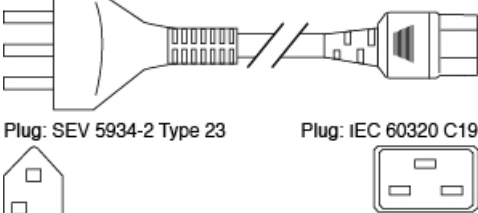
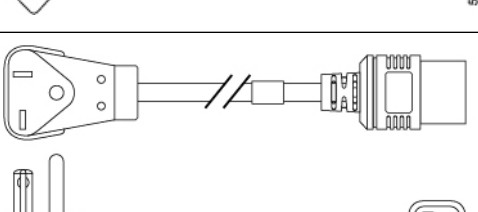
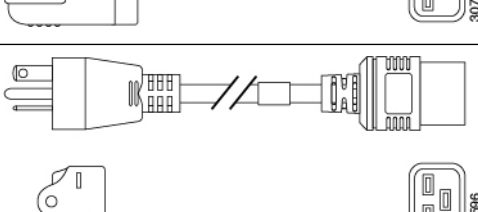
Power Cable Specifications

These sections show the power cables that you can order and use with this switch.

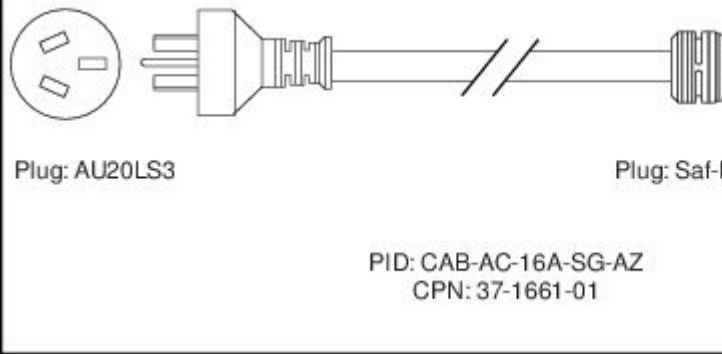
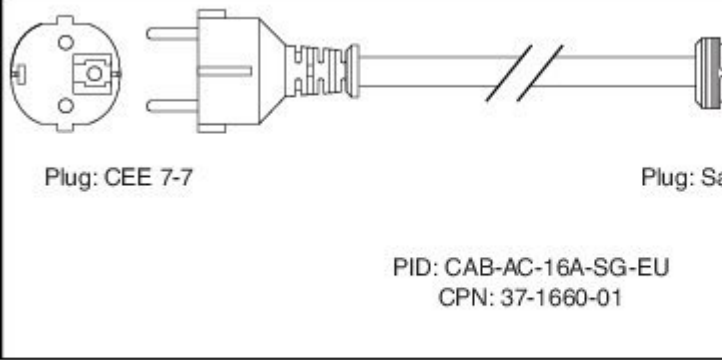
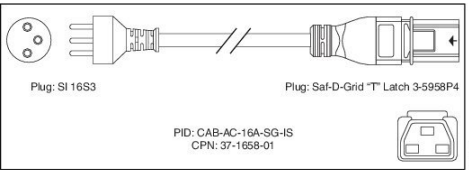
3-kW AC Power Cable Specifications

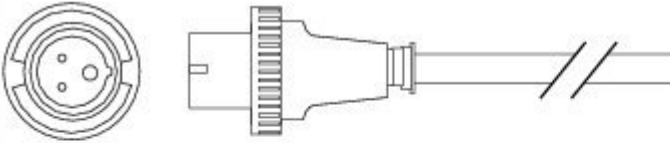
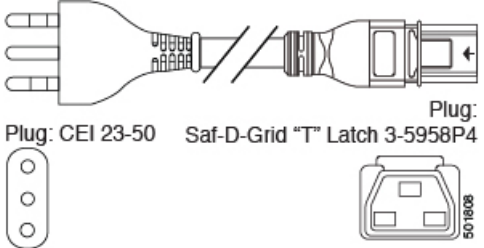
Locale	Power Cord Part Number	Cord Set Rating	Power Cord Illustration
Argentina	CAB-9K16A-ARG	16 A, 250 VAC	
Brazil	CAB-9K16A-BRZ	16 A, 250 VAC	
Korea	CAB-9K16A-KOR	16 A, 250 VAC	
South Africa	CAB-9K16A-SA	16 A, 250 VAC	TBD
Australia and New Zealand	CAB-AC-16A-AUS	16 A, 250 VAC	Plug: AU20S3 Plug: IEC 60320 C19
Peoples Republic of China	CAB-AC-16A-CH	16 A, 250 VAC	Plug: GB16C Plug: IEC 60320 C19

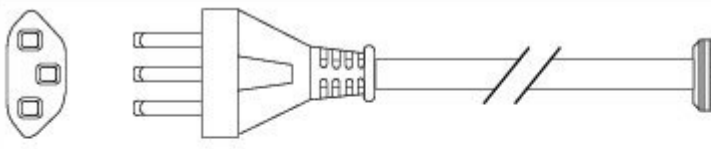
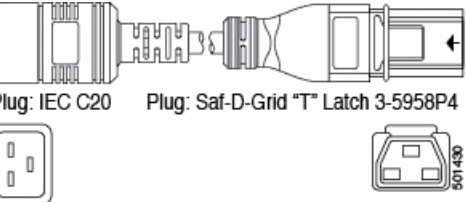
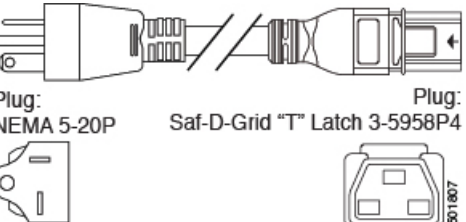
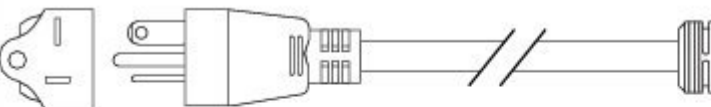
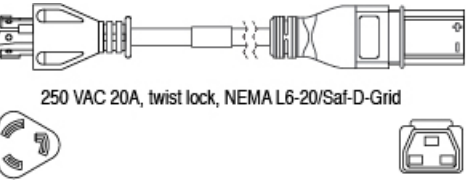
Locale	Power Cord Part Number	Cord Set Rating	Power Cord Illustration
Continental Europe	CAB-AC-2500W-EU	16 A, 250 VAC	 Plug: CEE 7/7 Plug: IEC 60320 C19 501422
India	CAB-C19-C20-IND	16 A, 250 VAC	 Plug: IEC 60320 C20 Plug: IEC 60320 C19 501428
India	PWR-CORD10-IND	16 A, 250 VAC	 Plug: IS16A3 Plug: IEC 60320 C19 504361
International	CAB-AC-2500W-INT	16 A, 250 VAC	 Plug: IEC 309 Plug: IEC 60320 C19 501423
Israel	CAB-AC-2500W-ISRL	16 A, 250 VAC	 Plug: SI16S3 Plug: IEC 60320 C19 501427

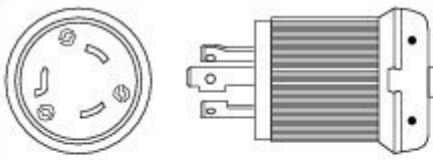
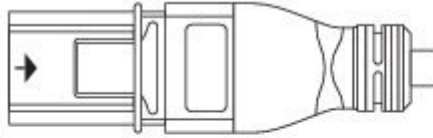
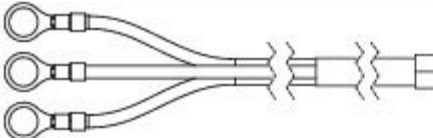
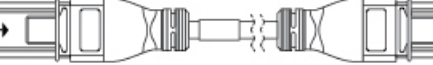
Locale	Power Cord Part Number	Cord Set Rating	Power Cord Illustration
Japan and North America (nonlocking) 200-240 VAC operation	CAB-AC-2500W-US1	16 A, 250 VAC	 Plug: NEMA 6-20 Plug: IEC 60320 C19
Japan and North America (locking) 200-240 VAC operation	CAB-AC-C6K-TWLK	16 A, 250 VAC	 Plug: NEMA L6-20 Plug: IEC 60320 C19
Power distribution unit (PDU)	CAB-C19-CBN	16 A, 250 VAC	 Plug: IEC 60320 C20 Plug: IEC 60320 C19
Switzerland	CAB-ACS-16	16 A, 250 VAC	 Plug: SEV 5934-2 Type 23 Plug: IEC 60320 C19
Taiwan	CAB-AC-C19-TW	16 A, 250 VAC (IEC60320 C19 to CNS 15767-1 & CNS 690)	
US	CAB-AC-STRT-C19US	16 A, 250 VAC (straight blade NEMA 6-20 plug)	
All except Argentina, Brazil, and Japan	NO-PWR-CORD	No power cord is included with order.	N.A.

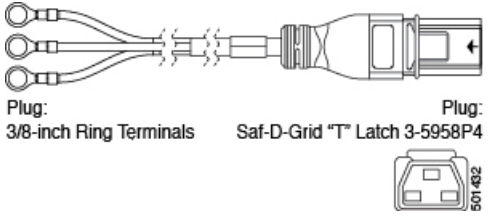
3-kW Universal AC/DC and 3-kW Dual Input Universal AC/DC Power Cable Specifications

Locale	Power Cord Part Number	Cord Set Rating	Power Cord Illustration
Australia and New Zealand	CAB-AC-16A-SG-AZ	16 A, 250 VAC (AU20LS3/Saf-D-Grid)	 Plug: AU20LS3 Plug: Saf-D-Grid PID: CAB-AC-16A-SG-AZ CPN: 37-1661-01
	CAB-AC-16A-SG-EU	16A, 250 VAC	 Plug: CEE 7-7 Plug: Saf-D-Grid PID: CAB-AC-16A-SG-EU CPN: 37-1660-01
Israel	CAB-AC-16A-SG-IS	16 A, 250 VAC	 Plug: SI 16S3 Plug: Saf-D-Grid "T" Latch 3-5958P4 PID: CAB-AC-16A-SG-IS CPN: 37-1658-01

Locale	Power Cord Part Number	Cord Set Rating	Power Cord Illustration
International/UK	CAB-AC-16A-SG-IN	16 A, 250 VAC	 Plug: IEC60309 PID: CAB-AC-16A-SG-I CPN: 37-1659-01
Italy	CAB-AC-16A-SG-IT	16 A, 250 VAC	 Plug: CEI 23-50 Saf-D-Grid "T" Latch 3-5958P4 Plug: 501808
South Africa	CAB-AC-16A-SG-SA	16 A, 250 VAC	 Plug: EL Plug: Saf-D- PID: CAB-AC-16A-SG-SA CPN: 37-1647-01

Locale	Power Cord Part Number	Cord Set Rating	Power Cord Illustration
Switzerland	CAB-AC-16A-SG-SW	16 A, 250 VAC	 Plug: SEV 5934-2 Plug: Saf-D-Grid "T" Latch 3-5958P4 PID: CAB-AC-16A-SG-SW CPN: 37-1654-01
North America	CAB-AC-20A-SG-C20	250 VAC 20 A	 Plug: IEC C20 Plug: Saf-D-Grid "T" Latch 3-5958P4 501430
North America	CAB-AC-20A-SG-US	16 A, 250 VAC	 Plug: NEMA 5-20P Saf-D-Grid "T" Latch 3-5958P4 Plug: Saf-D-Grid "T" Latch 3-5958P4 501807
North America (non locking) 200-240 VAC operation	CAB-AC-20A-SG-US2	250 VAC 20 A	 Plug: NEMA 6-20 Plug: Saf-D-Grid "T" Latch 3-5958P4 PID: CAB-AC-20A-SG-US2 CPN: 37-1657-01
North America	CAB-AC-20A-SG-US3	250 VAC 20 A	 250 VAC 20A, twist lock, NEMA L6-20/Saf-D-Grid 502586

Locale	Power Cord Part Number	Cord Set Rating	Power Cord Illustration
North America 277 VAC operation	CAB-AC-20A-SG-US4	277 VAC 20 A	 Plug: NEMA L7-20P PID: CAB-AC-20A-SG-US4 CPN: 37-1645-1
International, Saf-D-Grid/Saf-D-Grid	CAB-HV-25A-SG-IN1	400 VAC 20 A	 Plug: Saf-D-Grid "T" Latch 3-5958P4  Plug: Saf-D-Grid "T" Latch 3-5958P4 PID: CAB-HV-25A-SG-IN1 CPN: 37-1642-01
International, Ring Terminal source plug, Ring Terminal/Saf-D-Grid	CAB-HV-25A-SG-IN2	20A, 300 VAC/500 VDC	 Plug: 3/8-inch Ring Terminals Plug: Saf-D-Grid "T" Latch 3-5958P4 PID: CAB-HV-25A-SG-IN2 CPN: 37-1640-01
North America	CAB-HV-25A-SG-US1	277 VAC/ 240 VDC/ 380 VDC 25 A	 Plug: Saf-D-Grid "T" Latch 3-5958P4 Plug: Saf-D-Grid "T" Latch 3-5958P4  

Locale	Power Cord Part Number	Cord Set Rating	Power Cord Illustration
North America	CAB-HV-25A-SG-US2	277 VAC/ 240 VDC/ 380 VDC 25 A	 Plug: 3/8-inch Ring Terminals Plug: Saf-D-Grid "T" Latch 3-5958P4 501432
All except Argentina, Brazil, and Japan	NO-PWR-CORD	No power cord included with order	N.A.

3-kW DC Power Supply Power Cord Specifications

Each 3-kW DC power supply requires four customer-supplied power cables (two negative cables and two positive cables). We recommend using six gauge cables. Cisco supplies 6-gauge lugs for connections to the power supply. You supply the connectors that are required to connect the cables to the DC power source.



APPENDIX C

LEDs

- [Switch Chassis LEDs, on page 59](#)
- [Fan Module LEDs, on page 60](#)
- [Power Supply LEDs, on page 60](#)

Switch Chassis LEDs

The BCN, STS, and ENV, LEDs are located on the left side of the front of the switch. The port LEDs appear as triangles pointing up or down to the nearest port.

LED	Color	Status
BCN	Flashing blue	The operator has activated this LED to identify this switch in the chassis.
	Off	This switch is not being identified.
STS	Green	The switch is operational.
	Flashing amber	The switch is booting up.
	Amber	Temperature exceeds the minor alarm threshold.
	Red	Temperature exceeds the major alarm threshold.
	Off	The switch is not receiving power.
ENV	Green	Fans and power supply modules are operational.
	Amber	At least one fan or power supply module is not operating.
(port)	Green	Port admin state is 'Enabled', SFP is present and the interface is connected (that is, cabled, and the link is up).
	Amber	Port admin state is 'Disabled', or the SFP is absent, or both.
	Off	Port admin state is 'Enabled' and SFP is present, but interface is not connected.

LED	Color	Status
(GPS)	Green	GPS interface provisioned and ports are turned on. ToD, 1PPS, 10MHz are all valid.
	Off	Either the interface is not provisioned, or the ports are not turned on. ToD, 1PPS, 10MHz are not valid.

Fan Module LEDs

LED	Color	Status
Status	Green	The fan module is operational.
	Red	The fan module is not operational (fan is probably not functional).
	Off	Fan module is not receiving power.

Power Supply LEDs

The power supply LEDs are located on the right portion of the power supply.

Table 1: Power Supply LEDs

LEDs	Status
Off	The power supply is not functioning due to lack of input or output power. The input voltage is below the acceptable operating range. There is no 12 V standby from a parallel power supply unit.
Solid Amber	Failure (no output) or one of these possible conditions: • AC input below operating range, but 12V standby power is being supplied from a parallel power supply unit, • Over voltage, • Over current, • Over temperature, or • OTP due to fan failure.
Blinking Amber (1 Hz)	Power supply is operating but a warning condition has occurred—possibly one of these conditions: • High temperature, • High power, or • Slow fan.

LEDs	Status
Blinking Green (2 Hz)	Input is present and its voltage is within the operating range. The power supply is in standby mode due to PSON_L=High.
Blinking Green (1 Hz)	The power supply is Bootloading (startup mode). Note This applies only to NXA-PDC-2KW-PI and NXA-PDC-2KW-PE power supplies.
Solid Green	The power supply provides a regulated 12 V main output and a 12 V standby output.



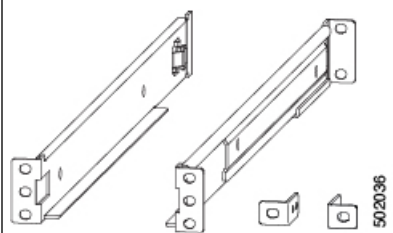
APPENDIX D

Additional Kits

- [Rack Mount Kit N9K-C9300-RMK](#), on page 63

Rack Mount Kit N9K-C9300-RMK

The following table lists and illustrates the contents for the 2-RU rack-mount kit (N9K-C9300-RMK).

Illustration	Description	Quantity
	Rack-mount kit • Bottom support rails (2) • Front-mount brackets (2)	1



APPENDIX E

Site Preparation and Maintenance Records

- [Site Preparation Checklist, on page 65](#)
- [Contact and Site Information, on page 66](#)
- [Chassis and Module Information, on page 67](#)

Site Preparation Checklist

Planning the location and layout of your equipment rack or cabinet is essential for successful switch operation, ventilation, and accessibility.

The table lists the site planning tasks. We recommend that you complete the tasks before you install the switch. Your completion of each task ensures a successful switch installation.

Planning Activity		Verification Time and Date
Space evaluation:		
	Space and layout	
	Floor covering	
	Impact and vibration	
	Lighting	
	Physical access	
	Maintenance access	
Environmental evaluation:		
	Ambient temperature	
	Humidity	
	Altitude	
	Atmospheric contamination	
	Airflow	

Planning Activity	Verification Time and Date
Power evaluation:	
Input power type	
Power receptacles	
Receptacle proximity to the equipment	
Dedicated (separate) circuits for power redundancy	
UPS for power failures	
Grounding: proper wire gauge and lugs	
Circuit breaker size	
Grounding evaluation:	
Data center ground	
Cable and interface equipment evaluation:	
Cable type	
Connector type	
Cable distance limitations	
Interface equipment (transceivers)	
EMI evaluation:	
Distance limitations for signaling	
Site wiring	
RFI levels	

Contact and Site Information

Use the worksheet to record contact and site information for the installation.

Contact person	
Contact phone	

Contact e-mail	
Building/site name	
Data center location	
Floor location	
Address (line 1)	
Address (line 2)	
City	
State/Province	
Contact person	
ZIP/postal code	
Country	

Chassis and Module Information

Use the worksheet to record information about the switch.

Contract number	
Chassis serial number	
Product Identification (PID) number	

Use the worksheet to record network-related information.

Switch IP address	
Switch IP netmask	
Hostname	
Domain name	
IP broadcast address	
Gateway/router address	
DNS address	

Use the worksheet to record information about the modules in the switch.

Module Slot	Module Type	Module Serial Number	Notes
Fan module 1			

Module Slot	Module Type	Module Serial Number	Notes
Fan module 2			
Fan module 4			
Power Supply 1			
Power Supply 2			
QSFP/QSFP-DD Tranceiver			
SFP/SFP+ Tranceiver			